

HORMONALY PRESS · 2026 A POCKET REFERENCE

A CLINICIAN'S POCKET GUIDE · 2026

Aesthetic & Regenerative Endocrinology

A clinician-focused reference to the skin–endocrine system: the physiology of endocrine aging, the organs of aging, and *evidence-graded interventions* through to the regenerative frontier.

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FIRST EDITION · 2026

Aesthetic & Regenerative Endocrinology

A pocket guide for the practitioner – optimizing the skin-endocrine system.

A concise, clinician-focused reference covering the physiology of endocrine aging, the decline of hormones and peptides over the lifespan, the organs of aging, and the evidence-graded interventions – peptides and bioregulators, hormone optimization, the gut, supplements and senotherapeutics, lifestyle and circadian medicine – through to the regenerative frontier of endocrine tissue replacement. With strength-of-evidence ratings and references throughout.

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How to use this guide

This pocket guide is built for the clinician who wants a clear, current picture of aesthetic and regenerative endocrinology — the optimization of skin, gut, and the endocrine axes for healthspan — without wading through a textbook. It moves in four parts: foundations, the organs of aging, interventions, and the regenerative frontier with a framework for practice. It is designed to be read in an afternoon or consulted at the point of care.

HOW EACH CHAPTER IS BUILT

- A one-line statement of the chapter's clinical idea, then focused prose with key terms in bold.
- Inline citations^[1] point to a numbered References block at the foot of each chapter. Numbering restarts at 1 in every chapter, so a citation always refers to its own chapter's list, never a running total.
- Strength-of-evidence pills (**GRADE A** **GRADE B** **GRADE C** **GRADE D**) mark how good the evidence is for the specific use described — not the elegance of the biology, nor its marketing.
- Callout boxes carry the practical takeaway — and, in every chapter, clinical pearls for both testing and therapy.
- A closing "three things to remember" strip distills each chapter to its core.

STRENGTH OF EVIDENCE AT A GLANCE



FROM ESTABLISHED TO SPECULATIVE, WITH REPRESENTATIVE INTERVENTIONS · SCHEMATIC, NOT TO SCALE

SCOPE & INTENT

Evidence grades reflect evidence quality for the specific indication discussed, not biological plausibility or mechanistic elegance. This field attracts genuine advances and considerable hype; the guide states plainly what is established, what is a useful adjunct, and what remains investigational or unapproved. It is a reference, not a protocol, and not medical advice for any individual — it informs clinical reasoning but does not replace specialist input, formal guidelines, or individualized judgment. Approvals, compounding rules, and the evidence base change quickly: verify against current sources, and confirm every citation against the primary literature, before acting.

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PART

Part I · Foundations

The biology the rest of the guide stands on – what regenerative endocrinology is, how the endocrine system ages, and the peptides and timing systems that recur throughout.

IN THIS PART

- What is regenerative endocrinology?
- The aging endocrine system
- Peptides & bioregulators
- Circadian & neuroendocrine timing

Chapter 1 · What is regenerative endocrinology?

Using endocrine biology to extend the years lived in good health — not merely to chase a number.

Regenerative and aesthetic endocrinology applies the tools of endocrinology — hormones, signaling peptides, nutrient-sensing pathways, and the cell biology of repair — to the central goal of modern preventive medicine: extending healthspan, the span of life spent free of chronic disease and disability, rather than lifespan alone.[1] The aesthetic dimension is not vanity; the skin is the most visible readout of systemic endocrine and metabolic health, and the same processes that age it age the vasculature, the brain, and the bone.

The field rests on a unifying insight from geroscience: aging is driven by a finite set of interacting hallmarks — among them genomic instability, epigenetic drift, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem-cell exhaustion, altered intercellular communication, and chronic inflammation.[2,3] Hormonal decline both reflects and accelerates several of these. That is why an endocrine lens is powerful — and why it must be disciplined.

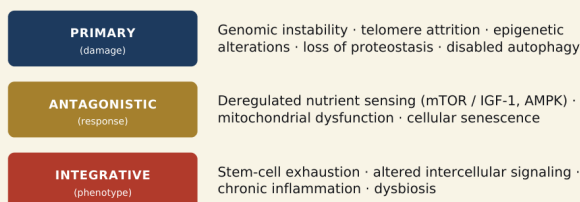


FIGURE 1.1 · THE HALLMARKS OF AGING — THE SUBSTRATE EVERY INTERVENTION IN THIS GUIDE ACTS UPON · SCHEMATIC

CHAPTER 1 • CONTINUED

A FIELD THAT ATTRACTS HYPE

Few areas of medicine generate more enthusiasm — or more unproven product — than "regenerative" and "anti-aging" practice. This guide holds a deliberate line: it states what is established and clinically actionable, what is emerging and reasonable to discuss, and what remains investigational or frankly speculative. Every major claim carries a strength-of-evidence rating, and the regulatory status of each agent is named plainly. The clinician's job is to separate biology that is real from a market that often outruns it.

CLINICAL DATA

~9 years	average lifespan–healthspan gap worldwide — years lived in poor health.[4]
1.0 → 2.1 billion	projected rise in adults aged 60+ between 2020 and 2050 (WHO).[5]
12 hallmarks	interacting aging processes in the 2023 update, expanded from the original nine — the substrate every intervention acts on.[3]

THREE THINGS TO REMEMBER

1. Regenerative endocrinology aims at healthspan — disease-free, functional years — using hormonal, metabolic, and cellular tools.
2. Aging is organized by interacting hallmarks; endocrine decline both signals and accelerates them.
3. The discipline is in the rating: state what is established, what is emerging, and what is investigational — and never blur them.

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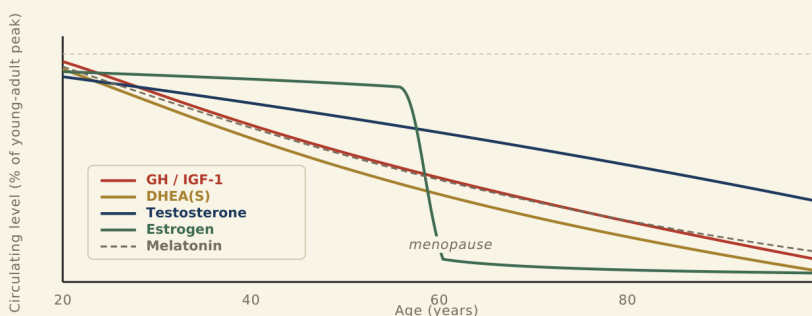
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Chapter 2 · The aging endocrine system

Each axis declines on its own schedule — the “-pauses” of endocrine aging.

Aging is not a uniform fade. Each endocrine axis follows a characteristic trajectory, and recognizing them is the first clinical skill in this field.[1] The somatopause is the progressive fall in growth-hormone pulse amplitude and IGF-1, beginning in the third decade and among the steepest of all.[2] Adrenopause is the parallel decline of DHEA and DHEA-sulfate, which by age 70–80 reach 10–20% of young-adult levels.[3] The andropause — better termed late-onset hypogonadism — is a gradual decline in total and (more steeply) free testosterone of approximately 0.5–2% per year, influenced by adiposity, illness, and rising SHBG.[4]

The female menopause is the exception: not a gentle slope but a cliff, with the loss of ovarian estradiol and progesterone over the perimenopausal transition, formally staged by STRAW+10.[5] Thyropause is subtler — TSH distributions shift upward with age, and modest subclinical changes are common and often benign in the very old. Melatonin amplitude and the robustness of cortisol's diurnal rhythm also blunt with age, themes returned to in Chapter 4.



Schematic; curves are illustrative, not to scale — rates and trajectories vary by individual

FIGURE 2.1 · CHARACTERISTIC ADULT DECLINE TRAJECTORIES OF MAJOR HORMONES — SCHEMATIC, NOT TO SCALE

CHAPTER 2 • CONTINUED

INTERPRETATION CAVEAT — A low level on a lab report is not automatically a deficiency to be "corrected." Reference ranges are age-stratified for a reason, and much endocrine decline is physiologic. Treatment is justified by symptoms plus confirmed biochemical deficiency against an appropriate range — not by a number alone.

CLINICAL DATA

~0.5–2%/yr	fall in total testosterone after ~age 35–40; free testosterone falls faster as SHBG rises.[4]
~10–20%	of young-adult DHEA-sulfate remaining by age 70–80.[3]
~51 years	median age of menopause; estradiol falls >90% across the transition.[5]
~14% / decade	fall in growth-hormone secretion through adult life after the third decade (somatopause).[2,6]

THREE THINGS TO REMEMBER

1. Each axis has its own clock: somatopause and adrenopause are steep and early; andropause is gradual; menopause is abrupt.
2. DHEA(S) falls to a small fraction of peak by old age; free testosterone declines faster than total with rising adiposity.
3. Physiologic decline is not disease — treat confirmed, symptomatic deficiency, not isolated low numbers.

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Chapter 3 · Peptides & bioregulators

What signaling peptides actually are — and a hard line between mechanism and proof.

Peptides are short amino-acid chains that act as signaling molecules: hormones (insulin, GLP-1), releasing factors (GHRH), and local repair and growth factors. The growth-hormone axis is the prototype — GHRH and ghrelin stimulate, somatostatin inhibits, and synthetic secretagogues (GHRH analogs like CJC-1295; ghrelin-receptor agonists like ipamorelin) act on these receptors to raise pulsatile GH.[1,2] The mechanistic plausibility is genuine; long-term human evidence in healthy aging is not.

The "bioregulator" tradition — short peptides such as epitagon (pineal) and thymalin (thymic), associated with Khavinson — proposes tissue-specific gene-regulatory effects.[3] The mechanistic literature is intriguing, but the human evidence is sparse, largely small and from a single research lineage, and falls short of supporting routine use. No adequately powered, multicenter randomized trials demonstrating improved healthspan or lifespan exist.

Regulatory status reflects June 2026 and may change; verify current FDA guidance before prescribing.

REGULATORY REALITY — VERIFY BEFORE PRESCRIBING (STATUS JUNE 2026)

As of June 2026, the U.S. FDA peptide-compounding landscape remains unsettled. Several popular peptides have appeared on, been withdrawn from, or are now under review in relation to the 503A bulk-substance lists: a 2023 round placed many in Category 2 (restricted), and in 2026 the FDA withdrew several from Category 2 and referred seven — BPC-157, KPV, TB-500, MOTS-c, DSIP (emideltide), semax, and epitagon — to its Pharmacy Compounding Advisory Committee (PCAC) for a 23–24 July 2026 meeting. Crucially, nomination, withdrawal from Category 2, or PCAC review is not FDA approval and does not by itself establish a settled lawful clinical compounding route; "research use only" is not a clinical loophole, and topical GHK-Cu (with limits) is the narrow Category 1 exception. Verify the current FDA 503A/503B lists, PCAC outcomes, and your state-board rules before clinical use.[4]

CHAPTER 3 • CONTINUED

A DECISION FRAME: IS THIS PEPTIDE USABLE?

CATEGORY • WHAT IT MEANS • CLINICAL STANCE

FDA-approved drug	Approved for a defined indication (e.g. tesamorelin, semaglutide) · Prescribe on-label; standard monitoring.
Pharmacy-compounded (503A/503B)	Made by a licensed pharmacy from a substance on the positive bulks list · Lawful only if the substance is currently listed and criteria are met — verify the live list.
Dietary supplement	Sold as a supplement (e.g. oral collagen peptides); no disease claims · Discuss as a supplement; quality and dose vary.
Investigational / under PCAC review	Nominated or being reviewed; not approved · Trials or IND only — review is not approval.
Research-use-only (RUO)	Labeled "not for human use"; no GMP/QC guarantee · Not a clinical route; impurity, dose, and contamination risk.

CLINICAL DATA

503A status	a 2023 round restricted several peptides; by 2026 several were withdrawn from Category 2 and seven (BPC-157, KPV, TB-500, MOTS-c, DSIP, semax, epitalon) were referred to PCAC for 23–24 Jul 2026. Review is not approval — no settled lawful compounding route yet.[4]
No benefit	no RCT shows functional or healthspan benefit from GH secretagogues in healthy aging adults — body-composition shifts have not translated into clinical gain, and carry metabolic downsides.[2]
Single-lineage	most bioregulator (epitalon) human data come from small, largely one-center studies.[3]
Plausible, not proven	the mechanistic rationale for signaling peptides is genuine, but adequately powered, multicenter healthspan or lifespan trials do not yet exist.
"RUO" is not a route	research-use-only material carries no GMP or quality guarantee and is not a lawful clinical pathway — impurity, dose, and contamination are real risks.

CHAPTER 3 • CONTINUED

THREE THINGS TO REMEMBER

1. Peptides are signaling molecules; mechanistic plausibility is common, long-term human proof in healthy aging is rare.
2. GH secretagogues act on the real GHRH/ghrelin axis but lack outcome evidence for "anti-aging" use.
3. The FDA 503A peptide landscape is unsettled — substances have moved on, off, and into PCAC review in 2026; nomination, withdrawal, or review is not approval, so no settled lawful compounding route exists yet. Check the current FDA status.

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Chapter 4 · Circadian & neuroendocrine timing

When a hormone is secreted matters as much as how much — and timing is modifiable.

The suprachiasmatic nucleus entrains a body-wide network of molecular clocks that gate metabolism, hormone secretion, and tissue repair across the 24-hour day.[1] Cortisol peaks near waking; growth hormone and most tissue repair are nocturnal and sleep-dependent; melatonin rises in darkness; insulin sensitivity is highest in the morning.[2] Aging blunts the amplitude of these rhythms — flatter cortisol curves, lower melatonin, fragmented slow-wave sleep — and that flattening is itself a driver of metabolic and immune decline.

Two levers are both potent and free. Light is the dominant zeitgeber: ordinary evening room light meaningfully suppresses and delays melatonin, degrading sleep quality and downstream repair.[3] The morning is the mirror image: stepping into outdoor light soon after waking — ideally within an hour — advances and stabilizes the central clock, the strongest daily anchor available.[4] Because the entraining signal is ocular — melanopsin-containing retinal cells that project to the SCN — the light must reach the eyes (go outside rather than rely on glass; never look straight at the sun). Letting it fall on the body and skin is a sensible bonus — the skin carries its own peripheral clocks, and morning time outdoors supports vitamin D and mood — but it does not replace ocular light, and it is no license for unprotected midday UV. Meal timing is the second: eating in alignment with the active phase — for example early time-restricted eating — improves insulin sensitivity and blood pressure even without weight loss.[5] Circadian alignment is the substrate on which every other intervention in this guide is built.

CLINICAL PEARL — Before adding any peptide, hormone, or supplement for "fatigue" or "poor recovery," fix the circadian basics first: consistent wake time, bright morning light, dim evening light, and a protected sleep window. Many presentations resolve here — at no cost and no risk.

CHAPTER 4 • CONTINUED

CLINICAL DATA

≥10,000 lux	outdoor morning light, versus ~100–500 lux indoors; get it into the eyes within about an hour of waking.[4]
0.3–0.5 mg	physiologic melatonin dose — far below the typical 3–10 mg in OTC products.[6]
~100 lux	ordinary evening room light is enough to suppress and delay melatonin.[3]
Weight-neutral	early time-restricted eating improved insulin sensitivity and blood pressure without weight loss.[5]

THREE THINGS TO REMEMBER

1. Hormone secretion is rhythmic; aging flattens rhythm amplitude, which itself harms metabolism and immunity.
2. Bracket the day with light: outdoor morning light into the eyes anchors the clock (body and skin exposure is a bonus, not a substitute), while dim evening light protects melatonin.
3. Eating earlier in the active phase improves insulin sensitivity independent of weight loss.

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PART

Part II · The Organs of Aging

Aging read organ by organ — skin as an endocrine and neuro-immuno-endocrine tissue, its glucocorticoid and estrogen biology, adipose, the gut, and the inflammatory-metabolic axis that links them.

IN THIS PART

- The skin as an endocrine organ
- The skin as a neuro-immuno-endocrine organ
- The endocrine atlas of skin
- The skin-glucocorticoid axis
- Estrogen and skin aging
- Endocrine disruptors, "clean beauty" & the skin
- Adipose tissue as an endocrine organ
- The gut-endocrine axis
- Inflammaging & metabolic aging

Chapter 5 · The skin as an endocrine organ

The most visible readout of systemic endocrine and metabolic aging.

Skin aging proceeds on two tracks. Intrinsic (chronological) aging thins the epidermis and dermis, reduces collagen by roughly 1% per year, fragments the elastic network, and lowers fibroblast output — changes first quantified decades ago.[1] Extrinsic aging, dominated by ultraviolet exposure, drives matrix metalloproteinase activation that degrades collagen faster than it is replaced, producing the coarse wrinkling and dyspigmentation of photoaging; glycation and smoking compound it.[2]

Crucially, the skin is not a passive recipient of hormones — it is a peripheral endocrine organ that expresses receptors for, and locally synthesizes, a broad range of signaling molecules, and performs its own steroidogenesis.[3,4] The table summarizes the major circulating inputs; the sections that follow add the detail and the locally produced peptides that the table cannot hold.

ENDOCRINE INPUT · CUTANEOUS ROLE

Estrogens	Collagen synthesis, dermal thickness, hyaluronan, barrier hydration, wound healing; menopausal loss drives thinning, dryness, lost elasticity
Androgens (T → DHT via 5 α -reductase)	Sebaceous size and sebum → acne, seborrhea, hirsutism, androgenetic alopecia; skin is itself a site of androgen metabolism
Glucocorticoids (cortisol)	Dermal atrophy, suppressed collagen, impaired healing, striae; local cutaneous HPA axis (Chapter 8)
Vitamin D (calcitriol)	Keratinocyte proliferation and differentiation, barrier, cutaneous immunity; synthesis begins in skin via UVB
DHEA, progesterone	Modest, largely precursor-driven roles in local steroidogenesis
Thyroid (T3 / T4)	Epidermal turnover, sebum and sweat, hair cycling; hypo → dry, coarse, cool skin; hyper → warm, moist skin
GH / IGF-1	Fibroblast proliferation, collagen, dermal thickness; IGF-1 raises sebum; insulin resistance → acanthosis nigricans
Melanocortins (α -MSH, ACTH)	POMC-derived; melanogenesis via MC1R; anti-inflammatory

CHAPTER 5 • CONTINUED

Prolactin, oxytocin	Cutaneous receptors; hair-follicle biology, sebaceous function, wound healing
PTHrP (local)	Regulates keratinocyte proliferation and differentiation

STEROID HORMONES

Estrogens are probably the single most influential class for skin quality: they stimulate fibroblast collagen synthesis, raise dermal thickness and hyaluronic acid content, support barrier hydration, and accelerate wound healing, so the post-menopausal estrogen drop drives much of the thinning, dryness, and lost elasticity of aging skin.[5,6] Androgens — testosterone and, more potently, DHT generated locally by 5 α -reductase — enlarge sebaceous glands and raise sebum, making them central to acne, seborrhea, hirsutism, and androgenetic alopecia; the skin is itself a site of androgen metabolism.[4] Glucocorticoid excess thins the dermis, suppresses collagen, impairs healing, and produces striae, through both systemic cortisol and a local cutaneous HPA-type axis — relevant to Cushing's and to chronic stress alike (Chapter 8). The hormonal form of vitamin D (calcitriol) governs keratinocyte proliferation and differentiation and modulates cutaneous immunity, and its synthesis begins in the skin under UVB;[7] DHEA and progesterone play more modest, partly precursor-driven roles.

THYROID HORMONES

T3 and T4 govern epidermal turnover, sebum and sweat secretion, and hair cycling. Hypothyroidism gives dry, coarse, cool skin and myxedema; hyperthyroidism gives warm, moist skin and, in Graves' disease, pretibial myxedema.[8]

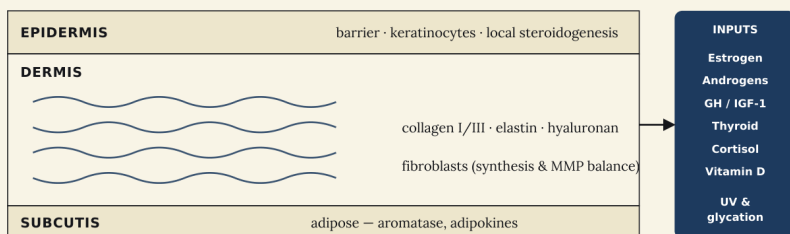
PEPTIDE AND PROTEIN HORMONES

Growth hormone and IGF-1 promote fibroblast proliferation, collagen synthesis, and dermal thickness; IGF-1 also stimulates sebaceous activity, part of why high-glycemic, high-insulin states track with acne,[9] while marked insulin resistance surfaces cutaneously as acanthosis nigricans. The melanocortins α -MSH and ACTH — both POMC-derived — drive melanogenesis through MC1R and are anti-inflammatory in skin.[3] Prolactin and oxytocin have cutaneous receptors with roles in hair-follicle biology, sebaceous function, and wound healing,[10,11] and locally produced PTHrP regulates keratinocyte proliferation and differentiation.[4]

CHAPTER 5 · CONTINUED

SKIN AS A SIGNALING TISSUE

A large amount of skin biology runs on paracrine and autocrine peptides rather than circulating hormones: EGF and FGF7/KGF for keratinocyte proliferation, TGF- β for fibroblast activation and scarring, VEGF for angiogenesis, and PDGF for wound repair.[12] Neuropeptides such as substance P and CGRP mediate neurogenic inflammation, itch, and vasodilation,[13] while the antimicrobial peptides cathelicidin (LL-37) and the defensins anchor barrier immunity – with LL-37 specifically implicated in rosacea.[14,15] Locally synthesized melatonin and β -endorphin add antioxidant and stress-response layers.[3]



The skin both responds to and synthesizes hormones – a peripheral endocrine organ

FIGURE 5.1 · SKIN ARCHITECTURE AND ITS HORMONAL INPUTS · SCHEMATIC, NOT TO SCALE

WHAT THE EVIDENCE SUPPORTS

Among topical agents, retinoids have the strongest dermatologic evidence for reversing measurable signs of photoaging and remain first-line GRADE A. For systemic menopausal hormone therapy, skin benefit is real but secondary – never an indication on its own.[6] Oral collagen peptides show modest improvements in hydration, elasticity, and wrinkle parameters in randomized trials, though most are small and industry-funded GRADE C,[16] and topical GHK-Cu has decades of mechanistic and small-trial support for collagen stimulation GRADE C.[17] The familiar "cosmetic peptides" borrow directly from this endogenous signaling: signal peptides such as palmitoyl pentapeptide-4 (Matrixyl) and copper tripeptide GHK-Cu target matrix and collagen, and neuromodulatory acetyl hexapeptide-8 (Argireline) softens expression lines – generally with modest, small-trial evidence GRADE C.[18]

CHAPTER 5 • CONTINUED

COSMETIC & AESTHETIC ENDOCRINOLOGY

Although not a formally recognized subspecialty, these threads are increasingly discussed together under the label cosmetic (aesthetic) endocrinology — a descriptive framing for a familiar clinical activity. In practice it means treating endocrine pathophysiology to improve the cosmetically bothersome findings patients present with — most prominently the hirsutism, androgenetic alopecia, and acne of PCOS and hyperandrogenism.[19,20] The skin is its organ, and a growing literature frames the skin explicitly as an endocrine tissue that both produces and responds to hormones.[21]

The estrogen–skin axis is the best-developed lever. Because sex steroids are small lipophilic molecules, they can be delivered topically for local effect with limited systemic exposure;[20] oral or topical estrogen slows — though it cannot reverse — post-menopausal skin aging, improving collagen, thickness, and hydration, and selective estrogen-receptor modulators (SERMs) are being explored to capture that benefit more selectively.[22,23] The axis also runs through stress: psychological stress, acting via the HPA axis and androgens, is associated with acne — a reminder that a cosmetic complaint is often a systemic readout.[24]

The lever cuts both ways. Because each of these findings is downstream of a hormonal driver, treating the endocrine cause is often the most durable cosmetic intervention — more so than a topical measure applied to the surface alone. Suppressing ovarian and adrenal androgen output (a combined oral contraceptive, spironolactone, or 5 α -reductase inhibition) improves the acne, seborrhea, and hirsutism of hyperandrogenism; restoring euthyroidism reverses the dry, coarse skin and diffuse shedding of hypothyroidism; and correcting hypercortisolism resolves the dermal thinning and easy bruising it drives. The discipline of the field is to read the skin as a presenting sign, identify the axis behind it, and treat that axis to its guideline target — rather than chase the visible finding in isolation.

CLINICAL PEARL — THE AESTHETIC EXAM IS A SYSTEMIC SCREEN —
Sudden skin thinning and easy bruising suggest cortisol excess; coarse, dry, cool skin suggests hypothyroidism; new acanthosis nigricans flags insulin resistance; rapid photoaging tracks UV, glycation, and smoking. Read the skin as a window on the whole endocrine system before reaching for a serum.

CHAPTER 5 • CONTINUED

CUTANEOUS SIGNS OF ENDOCRINE DISEASE — QUICK REFERENCE

SKIN FINDING • POINTS TO

Acanthosis nigricans	Insulin resistance / hyperinsulinemia (T2D, PCOS, obesity)
Hirsutism, persistent acne	Androgen excess (PCOS, CAH, androgen-secreting tumor)
Violaceous striae, easy bruising, thin skin	Cortisol excess (Cushing syndrome)
Dry, coarse, cool skin; periorbital puffiness	Hypothyroidism
Flushing	Carcinoid, pheochromocytoma, menopause, rosacea
Hair loss (diffuse or patterned)	Thyroid disease, androgen excess, telogen effluvium

REFER TO ENDOCRINOLOGY PROMPTLY IF

- Suspected Cushing syndrome (violaceous striae, central obesity, proximal weakness, new hypertension or diabetes).
- Rapidly progressive virilization (deepening voice, clitoromegaly, rapid hirsutism) — possible androgen-secreting tumor.
- Severe or rapidly progressive alopecia areata.
- Unexplained weight loss.
- Signs of an endocrine tumor (new galactorrhea, visual-field loss, a neck mass, refractory hypertension).

CLINICAL DATA

~30% in 5 yr	fall in skin collagen in early postmenopause, then ~2%/yr thereafter.[5]
2.5–10 g/d	oral collagen-peptide dose improving skin hydration and elasticity in RCTs.[16]
Topical GHK-Cu	decades of mechanistic and small-trial support for collagen synthesis.[17]

CHAPTER 5 • CONTINUED

THREE THINGS TO REMEMBER

1. Skin ages intrinsically (collagen loss, thinning) and extrinsically (UV-driven MMP collagen breakdown).
2. The skin is a full endocrine organ: it senses and synthesizes steroid, thyroid, and peptide hormones and runs local signaling — growth factors, neuropeptides, and antimicrobial peptides.
3. Evidence ladder: topical retinoids (A) > oral collagen peptides, topical GHK-Cu, and other cosmetic peptides (C); hormone-therapy skin benefit is secondary, never an indication.

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CHAPTER 5 • CONTINUED

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Chapter 6 · The skin as a neuro-immuno-endocrine organ

From a single endocrine tissue to an integrated sensor that tunes systemic homeostasis.

The previous chapter treated the skin as an endocrine organ. The fuller picture, consolidated in recent reviews, is that the skin is a neuro-immuno-endocrine (NIE) organ: a peripheral system that senses the environment, mounts a local hormonal and immune response, and — through hormones, neuropeptides, and primed immune cells entering the circulation, or through sensory nerves signaling the brain — helps adjust systemic homeostasis and allostasis.[1] Some authors order the triad differently (neuro-endocrine-immune), stressing that the immune compartment often delivers the final phenotype, but the unifying claim is the same: the skin computes environmental information and reports it inward.[2] This is not a fringe idea — the hypothesis that the skin is a self-contained neuroendocrine organ, articulated decades ago, has since been validated experimentally across cell types and species.[3] What follows is the mechanistic foundation, framed deliberately as biology rather than a menu of interventions.

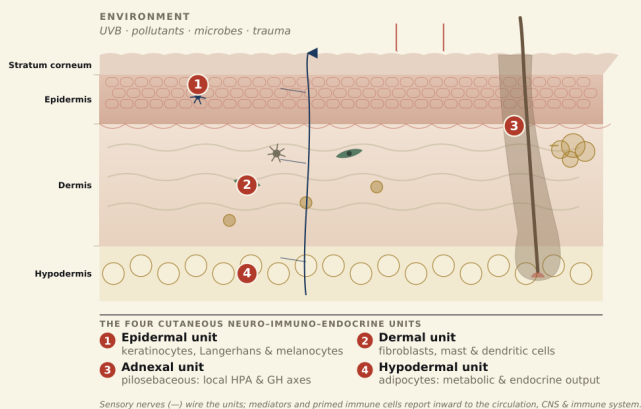


FIGURE 6.1 · THE FOUR CUTANEOUS IMMUNO-ENDOCRINE UNITS, COORDINATED BY SENSORY NERVES AND REPORTING TO THE CIRCULATION AND CNS · SCHEMATIC, NOT TO SCALE

CHAPTER 6 • CONTINUED

THE FOUR IMMUNO-ENDOCRINE UNITS

The skin's machinery is compartmentalized into four immuno-endocrine units — epidermal, dermal, hypodermal, and adnexal (the pilosebaceous unit and glands) — that can act alone or in concert.[1] Each contains resident cells with both hormone-making and immune capacity (keratinocytes, melanocytes, fibroblasts, sebocytes, adipocytes, mast cells, dendritic and Langerhans cells), wired together by sensory nerves that penetrate to the stratum corneum. Communication runs three ways: soluble factors between neighboring cells, rapid neural transmission, and release into the circulation that can reach distant organs.

A STEROIDOGENIC ORGAN, NOT JUST A CORTISOL SITE

Chapter 8 develops the cutaneous glucocorticoid axis; the broader fact is that human skin is a competent steroidogenic organ. Skin cells express CYP11A1, the rate-limiting enzyme that cleaves cholesterol to pregnenolone, and can build glucocorticoids, androgens, and estrogens locally — or activate precursors delivered from the blood.[4] Local interconversion uses familiar enzymes: 11 β -HSD1/2 for cortisol, 5 α -reductase for dihydrotestosterone, and 17 β -HSD for sex-steroid steps. Independent work shows keratinocyte-derived, de novo glucocorticoids actively police local immune tone rather than merely echoing systemic cortisol.[5]

CLINICAL DATA

Local cortisol	keratinocytes synthesize and activate cortisol on site; de novo glucocorticoids restrain cutaneous immune responses.[5]
Local DHT	5 α -reductase converts testosterone to DHT in the pilosebaceous unit — the substrate of acne, androgenetic alopecia, and hirsutism.
Read as foundation	much of the local-steroidogenesis literature is mechanistic (cell and ex-vivo); it explains biology, but is not a dosing rationale.

CHAPTER 6 · CONTINUED

SECOSTEROIDS — MESSENGERS OF THE SUN

In most individuals, cutaneous synthesis is the dominant natural source of vitamin D: in the skin, UVB converts 7-dehydrocholesterol to cholecalciferol. Recent work extends this to a family of photoproducts — lumisterol, tachysterol, and at least 18 CYP11A1-derived hydroxy-metabolites — that act not only on the vitamin D receptor but on ROR α/γ , AhR, LXR, and PPAR γ , with antiproliferative, pro-differentiation, and immunomodulatory effects.[6] The practical inference is modest but real: part of UVB's benefit is carried by signals oral vitamin D cannot reproduce — though most of these metabolites remain preclinical, and none of this licenses unprotected sun exposure.

LOCAL AXIS-EQUIVALENTS

Beyond the HPA axis, the skin runs truncated equivalents of other central axes. A cutaneous hypothalamic-pituitary-thyroid loop (local TRH and TSH, with deiodinases generating T3) modulates keratinocyte proliferation, the hair cycle, and mitochondrial function,[7] and a local somatotrophic loop (growth hormone and dermally produced IGF-1) influences dermal thickness, wound healing, and the hair follicle.[8]

LOCAL AXIS · CUTANEOUS MEDIATORS · REPRESENTATIVE ROLE

HPA (glucocorticoid)	CRH → POMC/ACTH → local cortisol; 11 β -HSD1/2 · Barrier, immune tone; atrophy with excess
Thyroid (HPT-like)	TRH, TSH; deiodinases D1/D2 → T3 · Keratinocyte proliferation, hair cycle, mitochondria
Somatotropic	GH; dermal IGF-1 · Dermal thickness, wound healing, hair
Sex steroids	CYP11A1, 5 α -reductase, aromatase · Sebum, hair growth, dermal collagen

POMC, MELANOCORTINS, AND THE OPIOID SYSTEM

The skin makes and processes POMC into ACTH, α -MSH, and β -endorphin, acting on melanocortin receptors MC1-MC5 to govern pigment, barrier strength, sebum, and local immunosuppression.[8] This biology has two clinical faces. The legitimate one is afamelanotide, an α -MSH analog approved to reduce phototoxicity in erythropoietic protoporphyria; the misused one is unregulated “tanning peptides” (melanotan II), sold online with no quality control GRADE D. Relatedly, UV-induced cutaneous β -endorphin can produce a genuine opioid-mediated drive to tan — a mechanism behind compulsive sun-seeking.[9]

THE NEURO AND IMMUNE DIMENSIONS — AND THE SYSTEMIC

CHAPTER 6 • CONTINUED

PAYOFF

On the neural side, sensory nerves releasing substance P and CGRP drive neurogenic inflammation and form the cutaneous arm of the brain-skin axis; the skin also makes melatonin and serotonin locally, with melatonin acting as a cutaneous antioxidant. On the immune side, skin-associated lymphoid tissue and keratinocyte cytokines do not stay local. The clearest healthspan link is the inflammatory skin “march”: chronic psoriatic inflammation is associated with insulin resistance and cardiometabolic disease, and the atopic march propagates allergic disease systemically.[10] Skin inflammation, in other words, is a plausible contributor to the same metabolic and vascular aging this guide targets elsewhere.

THE HAIR FOLLICLE: AN ENDOCRINE MINI-ORGAN

The pilosebaceous unit is the most autonomous of the adnexal structures — in effect a miniature endocrine organ. The human hair follicle runs functional peripheral equivalents of the central axes: it synthesizes cortisol and operates a local HPA-like loop,[11] and ex vivo it sustains a hypothalamic-pituitary-somatotropic equivalent (GHRH, GH, IGF-1) that paces its own growth.[12] It is also an immune-privileged site — normally shielded from immune attack — and collapse of that privilege drives alopecia areata, now treatable with JAK inhibitors that interrupt the responsible cytokine signaling (oral baricitinib is approved for severe disease GRADE A).[13]

Hormones set the follicle’s tempo. Thyroid hormone and prolactin tune the cycle and estrogen withdrawal at menopause thins hair, but the dominant clinical lever is androgenic: 5 α -reductase converts testosterone to DHT, which paradoxically miniaturizes genetically susceptible scalp follicles (androgenetic alopecia) while driving beard and body hair. This is why 5 α -reductase inhibitors (finasteride) and androgen-independent minoxidil are the mainstays of pattern hair loss, and why a sudden physiologic stressor or thyroid shift can tip follicles into telogen effluvium.

DISEASE MODEL • THE TRINITY IN ONE CONDITION

Rosacea integrates all three arms: immune sensing (TLR activation; cathelicidin cleaved to LL-37), an endocrine coordinator (CRH, inducible by even physiological UVB), and a neurovascular trigger (TRP channels, substance P, CGRP, mast-cell degranulation) — converging on flushing, inflammation, and angiogenesis.[2,14]

CHAPTER 6 • CONTINUED

THE SKIN MICROBIOME — THE FOURTH INTERFACE

These units do not operate in isolation: they sit in constant dialogue with a resident microbiome. Sebaceous sites are dominated by *Cutibacterium acnes*; moist and dry sites by *Staphylococcus epidermidis* and other commensals that educate cutaneous immunity, produce antimicrobial metabolites, and help hold the barrier in homeostasis GRADE B.[15] These organisms act as sentinels — in balance they restrain pathogens, but they can shift to opportunists, form biofilms, and tip into the dysbiosis seen in acne and atopic dermatitis GRADE B.[16]

With age the community itself changes: commensal diversity falls and *Cutibacterium* declines as sebum output drops — a microbial parallel to inflammaging that helps explain the drier, more reactive barrier of older skin.[16]

The clinical caveat is the usual one — the biology is real, but the “probiotic skincare” and live-bacterial-spray market runs well ahead of controlled outcome data GRADE C; treat the microbiome as an interface to respect (go easy on antiseptics and harsh surfactants), not yet a reliable therapeutic target.

Recent work sharpens this picture: metagenomic studies of aging skin show *Cutibacterium acnes* dominance falling and overall diversity rising, a shift that tracks measurably with reduced elasticity and more wrinkles,[17] and large multisite cohorts confirm the pattern across body sites — to the point that machine-learning models now estimate chronological age from the skin microbiome.[18]

The thread running through this chapter is integration: the skin does not merely host these systems side by side but runs them as one circuit, so a signal entering at any node — ultraviolet light, a shift in the microbiome, a surge of stress hormone, a breach of the barrier — propagates to the others and, often, inward to the body. That is why a single cutaneous complaint can carry endocrine, neural, immune, and microbial contributions at once, and why durable results usually come from addressing the upstream driver rather than any one downstream layer. The practical inference is modest but firm: read the skin as an information-rich readout of systemic state, respect its interfaces — barrier, microbiome, and circadian and stress inputs — before reaching for actives, and keep the mechanistic richness of this biology clearly separate from the far thinner evidence for intervening on it directly.

CHAPTER 6 • CONTINUED

THREE THINGS TO REMEMBER

1. The skin is a neuro-immuno-endocrine organ: four immuno-endocrine units (epidermal, dermal, hypodermal, adnexal) sense the environment and report inward via hormones, nerves, and primed immune cells.
2. It is a genuine steroidogenic and secosteroid factory (CYP11A1; vitamin D, lumisterol, tachysterol) and runs local HPA-, thyroid-, and GH/IGF-1-like axes — mechanistic foundation, not a dosing rationale.
3. Skin inflammation is not only local: the psoriatic and atopic “marches” link cutaneous immune activity to systemic cardiometabolic and allergic disease.

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CHAPTER 6 • CONTINUED

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Chapter 7 · The endocrine atlas of skin

A single-cell atlas of the skin's endocrine signaling network — and how it senesces with age.

Chapter 5 framed the skin as both a hormone target and a hormone producer. A cellular-resolution map now lets us infer that activity cell by cell — as transcriptomic predictions (which cells express the machinery to make or sense a hormone), not direct measurements of secretion or function. The Hormone Cell Atlas is a landmark transcriptomic resource; its cell-level findings represent predicted endocrine programs that require functional validation, and the clinical inferences below should be read in that light. The Hormone Cell Atlas profiled 379 hormone and receptor genes across roughly 14 million single cells from 47 human tissues — skin among them — and scored each cell type's predicted hormone-producing and hormone-receiving strength.[1] Two of its findings reframe how we should think about skin.

First, endocrine identity tracks cellular lineage more than tissue of origin, and the capacity is nearly universal: roughly 74% of cell types were predicted to produce, and 93% to respond to, at least one hormone.[1] Second, it follows that the skin — the body's largest organ, built from every major lineage (epithelial, mesenchymal, endothelial, adipocyte, immune, neural) in vast quantity — is plausibly one of the body's largest distributed endocrine signaling networks — an inference from transcriptomic prediction rather than direct measurement of secretion.[2] It is also endocrine-autonomous in ways few tissues are: it is the body's site of vitamin D synthesis,[3] and it runs a local, fully competent HPA-like axis — producing CRH, cleaving POMC to ACTH, α -MSH and β -endorphin, and synthesizing cortisol in situ.[4]

The scale argument is straightforward. At roughly 1.5–2 m² of surface and up to 15% of body weight once the subcutis is counted, the skin holds a far larger population of hormone-competent cells than the compact classical glands — the pituitary, adrenal, and thyroid are each measured in grams. What it lacks in concentrated, high-amplitude output it makes up in distributed scale, anatomical reach, and the autonomy to both make and respond to its own hormonal signals.[2]

THE SKIN'S ENDOCRINE CELL ATLAS

CHAPTER 7 • CONTINUED

SKIN COMPARTMENT • PRODUCES (REPRESENTATIVE) • SENSES
(KEY RECEPTORS)

Epidermis — keratinocytes	Vitamin D ₃ ; POMC peptides (α MSH, β -endorphin); CRH; cortisol; local steroid activation • VDR, glucocorticoid, estrogen (ER), androgen (AR), thyroid (TR), retinoid (RAR/RXR)
Dermis — fibroblasts	IGF-1; local growth factors; matrix-turnover signals • ER (collagen/hyaluronan), AR, GH/IGF-1, glucocorticoid, thyroid
Microvasculature — endothelium	Endothelin-1, adrenomedullin, neurotensin, prostaglandin D ₂ synthase, CTRP9 • Adrenomedullin, apelin, somatostatin (SSTR1)
Subcutis — adipocytes	Leptin, adiponectin, omentin, asprosin, resistin • Thyroid, GH, estrogen; depot/sex-specific AR & ESR2; NPY, BDNF
Resident immune cells	Visfatin (myeloid), thromboxane A ₂ (macrophage), PACAP (mast), MCH (Treg) • VDR (highest in dendritic cells)
Neuroectoderm — melanocytes & nerves	Melanin; substance P, CGRP (sensory nerves) • MC1R (α -MSH), adrenergic, neuropeptide receptors

Endothelial, adipocyte, and immune rows reflect atlas-derived, lineage-associated programs carried by the skin's resident cells of each type;[1] epidermal steroidogenesis, vitamin D, and neuropeptide rows draw on cutaneous endocrinology.[2–4]

SIGNALING PATHWAYS WORTH KNOWING

Vitamin D. The skin is the only organ that makes vitamin D de novo, converting 7-dehydrocholesterol to cholecalciferol under UVB; the atlas finds VDR broadly expressed, highest in epithelial and dendritic cells — a cellular basis for vitamin D's immunomodulatory role.[1,3] Local steroidogenesis. Genes for steroid activation and metabolism are widely distributed across epithelial tissues, letting target tissues such as skin shape steroid action locally[1] — the molecular correlate of the cutaneous HPA axis.[4]

CHAPTER 7 • CONTINUED

Adipose–skin crosstalk. The subcutis is an endocrine organ in its own right: its adipocytes secrete leptin and adiponectin and carry depot- and sex-specific receptor profiles, with androgen and ESR2 receptors enriched in subcutaneous fat.[1] Vascular–endocrine niche. Dermal endothelium is a source of endothelin-1 and adrenomedullin — a specialized vascular-endocrine compartment relevant to flushing, perfusion, and rosacea.[1] Neuro-immuno-endocrine. Mast cells express the stress neuropeptide PACAP and sensory nerves release substance P and CGRP, tying stress and neurogenic inflammation to skin disease.[1,4]

ENDOCRINE SENESENCE IN THE SKIN

The lineages this atlas maps do not age evenly. With chronological time and cumulative UV exposure, dermal fibroblasts, keratinocytes, and endothelial cells accumulate as senescent cells — growth-arrested but metabolically active, broadcasting a senescence-associated secretory phenotype (SASP) of cytokines, matrix metalloproteinases, and growth factors.[5] In skin the effect is visible: SASP-driven MMPs degrade dermal collagen, and the chronic IL-6/IL-8 tone is a cutaneous face of inflammaging (Ch. 13).

Senescence and endocrine signaling are coupled in both directions. Senescent cells rewire the local hormone and growth-factor milieu — the “altered intercellular communication” hallmark — blunting neighboring cells’ responsiveness to estrogen, IGF-1, and retinoids, while estrogen loss at menopause is itself associated with a heavier senescent burden and slower fibroblast turnover.[4,6] The practical lesson is that hormone optimization and senescence are not separate levers: correcting a true deficiency works partly by easing the senescent, pro-inflammatory state these endocrine cells fall into. Senolytic and senomorphic strategies to clear or quiet them (Ch. 17) are an active dermatologic frontier, though skin-specific human evidence is still early.

USING THE ATLAS • A CAVEAT

Skin-specific hormone and receptor lists live in the atlas’s supplementary tables (producing, S3; receiving, S6) and the browsable portal at hormonecellatlas.org.uk. [1] These are transcriptomic predictions — they flag the genes a cell expresses, not the quantity of processed, active hormone it secretes. Treat them as a map to confirm, not proof of secretion.

CHAPTER 7 • CONTINUED

CLINICAL PEARLS — TESTING & THERAPY — Testing: serum 25-OH vitamin D partly reflects cutaneous synthesis, so it falls with low UVB exposure and sunscreen use regardless of oral intake. Therapy: topical and transdermal routes exploit the skin's native receptors — retinoids (RAR), vitamin D analogs (VDR), estrogens (ER), glucocorticoids (GR) — but local steroid-metabolizing enzymes modify potency, so the cutaneous response need not mirror the systemic dose. The vitamin D analog calcipotriol is first-line topical therapy for plaque psoriasis — keratinocyte VDR signaling put to clinical use.[3]

THREE THINGS TO REMEMBER

1. The skin is the body's largest organ and, because endocrine activity tracks lineage and is near-universal, plausibly one of its largest distributed endocrine and peptide-signaling surfaces (a transcriptomic prediction, not a direct measurement).
2. Every compartment signals: keratinocytes (vitamin D, POMC, cortisol), fibroblasts (sex-steroid sensing), endothelium (endothelin/adrenomedullin), subcutis (leptin/adiponectin), immune and neural cells (PACAP, substance P).
3. The Hormone Cell Atlas predicts these programs from gene expression — a map to confirm at the protein level, not proof of secreted hormone.

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Chapter 8 · The skin–glucocorticoid axis

Cortisol is written on the skin – the “cortisol face,” and how to read and reset the diurnal rhythm.

The skin is not only a target of cortisol but a local glucocorticoid factory. Keratinocytes and fibroblasts express the glucocorticoid receptor, run a peripheral HPA-like axis (CRH → POMC → ACTH → cortisol), and regenerate active cortisol in situ through 11 β -HSD1 – an enzyme whose cutaneous activity rises with age, amplifying local glucocorticoid exposure just as the skin is least able to tolerate it.[1,2] Sustained glucocorticoid excess is visibly catabolic: it suppresses fibroblast collagen synthesis, thins the dermis and epidermis, delays barrier and wound repair, and produces the easy bruising, striae, and atrophy familiar from topical-steroid overuse.[3] Chronic psychological stress drives the same biology through the brain–skin axis, accelerating barrier dysfunction, inflammation, and visible aging.[4]

THE “CORTISOL FACE”

The popular term “cortisol face” describes facial puffiness, rounding, and fluid retention blamed on stress and high cortisol. Hold two truths at once. The genuine endocrine entity is the moon facies of Cushing’s syndrome – a rounded, plethoric face with dorsocervical and supraclavicular fat pads, central adiposity, wide purple striae, proximal muscle weakness, and easy bruising – a referable diagnosis, not a cosmetic state.[5] Most of what circulates online as “cortisol face,” however, is not Cushing’s: it is the lifestyle-and-fluid picture of poor sleep, alcohol, high salt, and inflammation. The clinician’s task is to reassure the many while screening the few who carry red flags.

CHAPTER 8 • CONTINUED

READING THE DIURNAL RHYTHM

Cortisol is a rhythm, not a single number. It surges 30–45 minutes after waking — the cortisol awakening response (CAR) — then falls across the day to a near-nadir at night.[6] Aging, chronic stress, and poor sleep flatten the curve and raise evening cortisol, a pattern tied to metabolic, immune, and cutaneous decline (Chapter 4). Two testing questions are distinct. To map the rhythm, a four-point diurnal salivary cortisol (waking, +30 min, afternoon, bedtime) is the practical tool. To screen for true hypercortisolism when red flags are present, use late-night salivary cortisol, 24-hour urinary free cortisol, or a 1 mg overnight dexamethasone-suppression test — and refer to endocrinology.[5] Consumer “cortisol panels” (random serum or hair cortisol) are poorly validated for the worried-well; resist over-testing.

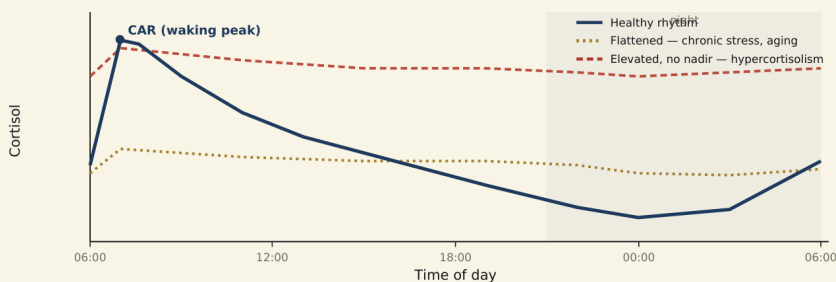


FIGURE 8.1 · THE DIURNAL CORTISOL RHYTHM — A HEALTHY MORNING PEAK (CAR) AND NIGHT-TIME NADIR, VERSUS THE FLATTENED PATTERN OF CHRONIC STRESS AND AGING AND THE ELEVATED, NADIR-LESS PATTERN OF HYPERCORTISOLISM · SCHEMATIC, NOT TO SCALE

CHAPTER 8 • CONTINUED

RESETTING THE RHYTHM

The levers that sharpen a healthy cortisol curve are behavioral and well evidenced GRADE B. Anchor the morning: a consistent wake time and bright morning light strengthen the CAR, while dim evening light and a protected 7–9-hour sleep window lower nocturnal cortisol (Chapter 4). Regular moderate exercise lowers resting cortisol, whereas chronic over-training and late high-intensity sessions can raise it. Stress-regulation practices — mindfulness, paced breathing, time in nature — measurably reduce cortisol output GRADE B.[7] Limit late caffeine and alcohol, and protect the barrier and limit UV, since ultraviolet exposure itself activates the cutaneous stress axis.[1]

CLINICAL PEARL · RHYTHM, NOT NUMBER — Treat the curve, not a lone value. A flattened, evening-heavy rhythm in a stressed patient is a lifestyle target, not a disease, and the “cortisol face” is usually fluid, sleep, and inflammation. But proximal myopathy, wide purple striae, unexplained bruising, or rapid central weight gain are red flags for Cushing’s syndrome and warrant formal screening and referral — not a cortisol supplement or a wellness panel.

CLINICAL DATA

+~50% / 30–45 min

normal cortisol awakening response after waking; a flattened curve tracks stress and poor sleep.[6]

Screen if red flags

late-night salivary cortisol, 24-h urinary free cortisol, or 1 mg overnight dexamethasone test for suspected Cushing syndrome.[5]

11 β -HSD1 $\uparrow$ with age

raises locally regenerated cortisol in skin, compounding dermal thinning and slow repair.[2]

Behavioral levers

morning light, sleep 7–9 h, exercise, and stress regulation lower cortisol output.[7]

THREE THINGS TO REMEMBER

1. The skin makes and responds to cortisol (a local HPA axis with 11 β -HSD1); chronic glucocorticoid excess thins skin, slows repair, and ages it.
2. Distinguish the lifestyle-and-fluid “cortisol face” from the moon facies of Cushing’s syndrome — screen red flags (purple striae, proximal myopathy, bruising) and refer.
3. Cortisol is a rhythm: map it with four-point salivary testing, and reset it with morning light, sleep, exercise, and stress regulation — not supplements.

CHAPTER 8 • CONTINUED

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Chapter 9 · Estrogen and skin aging

The hormone whose withdrawal most visibly ages the skin — and what the evidence supports doing about it.

If one hormone dominates the visible aging of women's skin, it is estrogen. Its decline at menopause opens a brief, steep, and largely quantifiable phase of skin aging that overlies the slow background of chronological change — and much of it can be partly reversed by restoring estrogen, though, as below, skin benefit is never on its own a reason to prescribe.[1]

WHAT ESTROGEN DOES IN THE SKIN

Skin is a broad estrogen target. It expresses estrogen receptors — predominantly ER β , with ER α and the membrane receptor GPER — on keratinocytes, fibroblasts, and hair-follicle and sebaceous cells. Through them estrogen drives fibroblast synthesis of type I and III collagen and hyaluronic acid, maintains dermal thickness and water content, supports the barrier, speeds wound healing, and modulates pigmentation.[1,2] Withdrawal of that signal is what leaves estrogen-deficient skin thinner, drier, less elastic, and slower to heal.

THE MENOPAUSE INFLECTION

The decline is not gradual. Skin collagen falls fastest in the first postmenopausal years — on the order of 30% within the first five years, then roughly 2% per year — so dermal aging tracks years since menopause more tightly than chronological age.[1,3] Elasticity and hydration fall on a similar curve, which is why the perimenopausal transition is so often when patients first notice an abrupt change.

Two features make this phase clinically distinctive. First, it is partly separable from photoaging: where ultraviolet damage concentrates on sun-exposed surfaces and coarsens them, estrogen-related change is more uniform — generalized thinning, dryness, and loss of firmness across exposed and covered sites alike. Second, it is most responsive to intervention early, while the dermal architecture is still relatively intact, which is why the perimenopausal window is when counseling on sun protection, barrier care, and — where indicated for other reasons — hormone therapy has the most to work with. Recognizing the pattern reframes the "sudden aging" patients report as a predictable, time-limited inflection rather than an inevitable slide.

CHAPTER 9 • CONTINUED



FIGURE 9.1 • DERMAL COLLAGEN AFTER MENOPAUSE: A STEEP EARLY DECLINE, THEN A SLOWER LOSS • SCHEMATIC, NOT TO SCALE

SYSTEMIC HORMONE THERAPY AND THE SKIN

Menopausal hormone therapy measurably improves skin collagen content, thickness, elasticity, and hydration in controlled studies (GRADE B).[4] But this is a secondary benefit, not an indication: hormone therapy is prescribed for vasomotor symptoms and bone protection, in appropriate women, under the timing and risk rules set out in Chapter 14 — skin improvement is a welcome bystander effect, never the reason to start.[4]

TOPICAL ESTROGENS, SERMS, AND "ESTROGEN-LIKE"

COSMECEUTICALS

Topical estrogen creams — estradiol, and the weaker estriol (also used vaginally for GSM) — improve dermal markers in small studies, but they are absorbed systemically, are not approved as cosmetic anti-aging agents, and should not be used off-label for wrinkles (GRADE C).[5] Selective estrogen-receptor modulators (tamoxifen, raloxifene) act as agonists in some tissues and antagonists in others, with mixed and sometimes adverse skin effects; they are not skin therapies.[5] Plant "phytoestrogens" such as genistein are weak, and the cosmetic evidence for them is thin.[2]

CHAPTER 9 • CONTINUED

VULVOVAGINAL TISSUE: THE ONE CLEAR ESTROGEN INDICATION

The clearest setting in which giving estrogen to tissue is both effective and appropriate is the genitourinary syndrome of menopause (GSM) — the vulvovaginal dryness, irritation, painful sex, and urinary symptoms of local estrogen loss, affecting a large fraction of postmenopausal women. Low-dose vaginal estrogen is first-line for moderate-to-severe GSM, with strong evidence, minimal systemic absorption, and no need for an added progestogen (GRADE A); vaginal DHEA and oral ospemifene are alternatives.[6] GSM is under-recognized and very treatable — worth asking about directly rather than waiting for the patient to raise it.

CLINICAL DATA

~30% / 5 yr	dermal collagen lost in the first five post-menopausal years, then ~2%/yr.[3]
Skin = bystander	systemic hormone therapy improves skin, but skin benefit is never an indication to prescribe it.[4]
Vaginal estrogen	first-line for moderate-severe GSM; minimal systemic absorption, no progestogen needed.[6]

The clinical backdrop matters: the Women's Health Initiative reset the risk-benefit conversation around menopausal hormone therapy and remains the reference point for counseling,[7] while more recent meta-analytic data show that MHT increases skin collagen, thickness, and elasticity and reduces wrinkling — real effects, but skin benefit is a secondary consideration, never an approved indication.[8]

THREE THINGS TO REMEMBER

1. Estrogen maintains dermal collagen, thickness, hydration, and healing (mainly via ER β); its menopausal loss drives a steep, early phase of skin aging that tracks years-since-menopause.
2. Systemic hormone therapy improves skin, but that is secondary — prescribe it for symptoms and bone under guideline rules (Chapter 14), never for skin alone.
3. The one clear estrogen-for-tissue indication is GSM: low-dose vaginal estrogen is first-line, effective, and minimally absorbed.

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Chapter 10 · Endocrine disruptors, "clean beauty" & the skin

What gets absorbed through the skin – separating endocrine evidence from marketing.

Endocrine-disrupting chemicals (EDCs) are exogenous compounds that interfere with hormone synthesis, transport, signaling, or metabolism – often by mimicking or blocking estrogen, androgen, or thyroid receptors.[1,2] Because the skin is a large absorptive surface in daily contact with personal-care products, cosmetics are one relevant route of exposure. The classes that recur in this conversation are well characterized: phthalates (fragrance carriers and plasticizers), with anti-androgenic signals;[3] parabens (preservatives), weakly estrogenic;[4] UV filters such as oxybenzone; triclosan; and certain bisphenols and PFAS. The mechanistic concern is genuine, and biomonitoring shows exposure is near-universal at the population level.

CLASS (EXAMPLE) · WHERE IT SHOWS UP · ENDOCRINE SIGNAL

Phthalates	Fragrance ("parfum"), nail products, plasticizers · Anti-androgenic
Parabens	Preservatives in creams and cosmetics · Weakly estrogenic
Oxybenzone (BP-3)	Chemical sunscreens · Weak estrogenic/anti-androgenic in vitro; absorbed systemically
Triclosan	Antibacterial products, some cosmetics · Thyroid / estrogenic signals
Bisphenols, PFAS	Packaging; some long-wear cosmetics · Estrogenic; metabolic / thyroid

CHAPTER 10 • CONTINUED

WHAT THE EVIDENCE DOES AND DOES NOT SHOW

The honest position sits between dismissal and alarm. Absorption is measurable: under maximal-use conditions, FDA studies found several chemical sunscreen filters — oxybenzone among them — reaching plasma concentrations above the threshold that triggers formal safety testing, though absorption is not the same as demonstrated harm.[5] At the same time, much of the toxicology is in vitro, animal, or high-dose, and human clinical endpoints from cosmetic-level exposure remain uncertain, with low-dose and mixture effects still debated.[6] Professional bodies therefore counsel precaution weighted toward vulnerable windows — pregnancy, infancy, and puberty — rather than blanket avoidance.[1]

"CLEAN BEAUTY" — USEFUL CORE, MARKETING SHELL

"Clean beauty" is a marketing term, not a regulated standard: the FDA does not define "clean" or "natural," and — even under the 2022 Modernization of Cosmetics Regulation Act, which added facility registration, adverse-event reporting, and safety substantiation — most cosmetics and ingredients still reach the market without premarket approval.[7] "Chemical-free" is a misnomer, and "natural" does not mean safe — botanical extracts are common sensitizers. The defensible core of the movement is narrow and reasonable: reduce unnecessary exposure to a few well-characterized disruptors, particularly fragrance and certain preservatives, and especially around pregnancy. The trap is letting EDC anxiety crowd out the interventions that actually protect the skin.

CLINICAL PEARL • COUNSEL PROPORTIONATELY — Keep the certainties in front. Daily sun protection and retinoids have strong evidence for skin health; harm from cosmetic-level EDCs is plausible but unproven. If a patient is worried, the high-yield moves are simple: favor fragrance-free products, consider mineral (zinc/titanium) sunscreens over oxybenzone — but keep using sunscreen — and tighten avoidance during pregnancy. Treat ingredient databases as starting points, not verdicts, and steer patients away from both greenwashing and chemophobia.

IN PREGNANCY & PRECONCEPTION — PROPORTIONATE, NOT

CHAPTER 10 • CONTINUED

ALARMIST

Offer a few low-effort swaps (fragrance-free products, less canned or ultraprocessed food, ventilation when using nail or hair products) rather than a long avoidance list. Keep proportion: the evidence is largely associative, so lead with the high-certainty wins — not smoking, sun protection, nutrition, sleep — over fear of trace cosmetic exposures. Aim for "better, not perfect"; avoidance spirals and anxiety are their own harm.

CLINICAL DATA

Above threshold	oxybenzone and other chemical UV filters reached plasma levels above the FDA testing threshold under maximal use — absorption, not proven harm.[5]
Unregulated terms	"clean," "natural," and "chemical-free" have no FDA definition; cosmetics rarely face premarket approval.[7]
Precautionary window	pregnancy, infancy, and puberty are where reducing EDC exposure has the clearest rationale.[1]
Certain > speculative	sun protection and retinoids outweigh speculative EDC harms; if concerned, choose mineral sunscreen but keep using it.

THREE THINGS TO REMEMBER

1. EDCs (phthalates, parabens, some UV filters) can mimic or block hormone receptors; the skin is a real route of exposure, but human harm from cosmetic-level use is unproven.
2. "Clean beauty" is unregulated marketing — "natural" is not "safe" and "chemical-free" is meaningless; reduce unnecessary exposure rather than chase labels.
3. Counsel proportionately: certain wins (sunscreen, retinoids) first; precautionary avoidance is highest-yield in pregnancy; mineral filters are a reasonable swap, but never skip sunscreen.

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Chapter 11 · Adipose tissue as an endocrine organ

The body's largest hormone-secreting organ — a metabolic command center, and a quiet driver of skin and metabolic aging.

For most of the twentieth century, fat was treated as inert ballast. It is now recognized as the body's largest endocrine organ by mass, secreting dozens of adipokines that regulate appetite, insulin sensitivity, immunity, reproduction, and vascular tone.[1] The turning point came in 1994 with the cloning of leptin — the product of the ob (LEP) gene — which proved that adipocytes signal directly to the brain.[2] Crucially, adipose tissue is not one organ but several: subcutaneous, visceral, and a thin dermal layer wrapped around the skin's appendages, each with distinct behavior.

THE MAJOR ADIPOKINES

Leptin signals energy sufficiency to the hypothalamus, suppressing appetite and permitting reproduction and thyroid activity; in obesity its levels are high, but the brain becomes leptin-resistant, so the satiety signal fails.[2] Adiponectin is the mirror image — insulin-sensitizing and anti-inflammatory through AMPK, it is paradoxically low in obesity, and that fall tracks with insulin resistance, dyslipidemia, and cardiovascular risk.[3] Resistin links adiposity to insulin resistance clearly in rodents, though its role in humans remains debated. Overlaid on all of this, an enlarged, stressed fat mass recruits macrophages and secretes TNF- α and IL-6 — the molecular bridge from excess adiposity to the chronic inflammation and insulin resistance of inflammaging (Chapter 13).

ADIPOKINE · PRINCIPAL ACTION · DIRECTION IN OBESITY/AGING

Leptin	Satiety signal to hypothalamus; energy expenditure · High, with central resistance
Adiponectin	Insulin-sensitizing, anti-inflammatory (AMPK) · Low
Resistin	Insulin resistance (clear in rodents, debated in humans) · High
TNF- α / IL-6	Pro-inflammatory; drive insulin resistance and inflammaging · High

DEPOT MATTERS: VISCERAL, ECTOPIC, AND THE PERSONAL FAT

CHAPTER 11 • CONTINUED

THRESHOLD

Where fat sits matters more than how much there is. Subcutaneous fat is a relatively safe sink; visceral and ectopic fat — in liver, muscle, pericardium, and pancreas — are metabolically toxic, driving insulin resistance and secreting the inflammatory adipokines above. The personal fat threshold concept explains why two people at the same BMI can differ so sharply: metabolic disease appears when an individual's safe storage capacity is exceeded and lipid spills into ectopic sites, not at any universal weight.

ADIPOSE AND THE SKIN

The skin carries its own fat. Dermal white adipose tissue (dWAT) is a distinct adipocyte layer in the reticular dermis that is far more than insulation: it cycles with the hair follicle, fuels wound healing, and mounts an active antimicrobial defense — reactive adipogenesis releases cathelicidin to fight *Staphylococcus aureus*.^[4] With age, dWAT becomes dysfunctional: its signaling shifts, hair-follicle support wanes, and a pro-fibrotic adipocyte-myofibroblast transition contributes to scarring and the loss of dermal resilience.^[5] At the same time, loss of subcutaneous facial fat is one of the least-appreciated drivers of the aged face — deflation and descent of discrete fat compartments, not just wrinkles, produce the hollowing that fillers and fat grafting aim to reverse.

THE GLP-1 ERA

Incretin therapy has made adipose biology unavoidable. GLP-1 and dual GIP/GLP-1 agonists (Chapter 12) shrink fat mass and improve the adipokine profile — raising adiponectin and lowering inflammatory tone, with cardiovascular-outcome benefit now shown in large trials such as SELECT^[6] — but rapid, large-volume fat loss also depletes the subcutaneous and dermal fat that gives the face its youthful contour, the phenomenon popularly called "Ozempic face." It is a vivid reminder that adipose tissue is not only a metabolic liability but a structural and endocrine asset; the aim of regenerative practice is the right fat, in the right place, signaling correctly.

CLINICAL DATA

1994	leptin cloned — the discovery that reframed fat as an endocrine organ. ^[2]
Low in obesity	adiponectin falls as fat mass rises, tracking insulin resistance and cardiovascular risk. ^[3]
Depot > mass	visceral and ectopic fat drive metabolic disease; subcutaneous fat is comparatively safe.

CHAPTER 11 • CONTINUED

THREE THINGS TO REMEMBER

1. Adipose tissue is an endocrine organ: leptin (satiety; high but resistant in obesity) and adiponectin (insulin-sensitizing; low in obesity) are the anchor hormones.
2. Fat distribution beats fat quantity – visceral and ectopic fat are the metabolically dangerous depots.
3. Skin has its own fat: dermal white adipose tissue defends and repairs, and subcutaneous facial-fat loss (including from GLP-1 therapy) is a major driver of the aged face.

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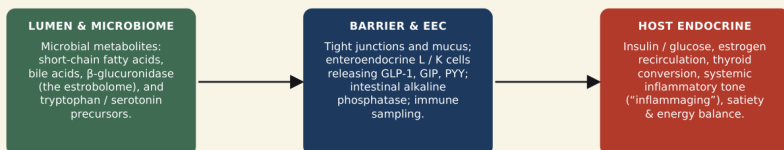
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Chapter 12 · The gut–endocrine axis

By cell number, the body's largest endocrine organ — and the gateway to inflammaging.

The gut is the body's largest endocrine surface. Enteroendocrine L- and K-cells secrete incretins — GLP-1, GIP, and PYY — that govern insulin release, gastric emptying, and satiety; this physiology underlies the most consequential metabolic drugs of the decade.[1] The GLP-1 receptor agonist semaglutide and the dual GIP/GLP-1 agonist tirzepatide produce double-digit weight loss with broad cardiometabolic benefit GRADE A, repositioning the gut hormones at the center of healthspan medicine.[2,3]

The gut is also an endocrine interface. The microbiome modulates host hormones — most strikingly through the estrobolome, the bacterial β -glucuronidase activity that deconjugates and recirculates estrogens, influencing systemic estrogen exposure.[4] When the barrier fails, microbial products such as lipopolysaccharide cross into the circulation, driving the low-grade, systemic inflammation ("metabolic endotoxemia") that links a leaky gut to insulin resistance and inflammaging.[5]



EEC = enteroendocrine cells. A leaky barrier shifts the host toward a pro-inflammatory, insulin-resistant state.

FIGURE 12.1 · THE GUT AS AN ENDOCRINE INTERFACE BETWEEN MICROBIOME, BARRIER, AND HOST · SCHEMATIC

CLINICAL PEARL

CLINICAL PEARL — A GLP-1 agonist is an endocrine, not merely a "weight-loss," intervention — it should be prescribed and monitored as one, with attention to lean-mass preservation (protein and resistance training), gastrointestinal tolerance, and the established cautions. The regenerative payoff of gut health is mostly indirect: lower inflammatory tone and better insulin sensitivity.

CHAPTER 12 • CONTINUED

CLINICAL DATA

~14.9%	mean body-weight loss with semaglutide 2.4 mg at 68 weeks (STEP-1).[2]
~21%	mean body-weight loss with tirzepatide 15 mg at 72 weeks (SURMOUNT-1).[3]
1–2 min	half-life of native GLP-1 (rapid DPP-4 breakdown) — why long-acting analogs were engineered.[1]
Estrobolome	gut bacterial β -glucuronidase deconjugates and recirculates estrogen, so dysbiosis can shift systemic estrogen exposure.
Barrier → inflammaging	a leaky gut barrier drives metabolic endotoxemia and low-grade inflammation — a mechanistic link from gut health to systemic and skin aging.

GLP-1 BEYOND METABOLISM: THE SKIN

The incretin story now reaches the skin. GLP-1 receptors are expressed on dermal fibroblasts and adipose-derived stromal cells, and beyond glucose and weight, GLP-1 receptor agonists show anti-inflammatory and immunomodulatory effects: small studies report improvement in psoriasis and other inflammatory dermatoses and enhanced (especially diabetic) wound healing GRADE C.[6] The trade-off is cosmetic and now familiar — rapid, large-magnitude weight loss depletes facial and dermal fat, producing the volume loss, laxity, and periorbital hollowing popularly called "Ozempic face" — most pronounced in older patients who have already lost collagen and elastin; telogen effluvium can follow the same rapid loss GRADE B.[7] Whether GLP-1 also acts directly on skin cells independently of weight is an open, actively studied question. The practical counsel: pace the loss, prioritize protein and resistance training, and raise the aesthetic trade-offs before starting therapy — the skin effects are emerging, not settled.

COUNSELING THE GLP-1 PATIENT — SKIN & BODY COMPOSITION

- Defend lean mass: adequate protein (about 1.2–1.6 g/kg/day) plus resistance training through the loss phase.
- Pace matters — slower titration and a gradual rate of loss reduce both lean-mass and facial-volume loss.
- Pre-empt the aesthetics: discuss facial-volume loss and laxity ("Ozempic face") before starting, especially in older patients.
- Telogen effluvium can follow rapid loss — usually self-limited; reassure and reassess.

CHAPTER 12 • CONTINUED

THREE THINGS TO REMEMBER

1. By enteroendocrine-cell number the gut is the body's largest endocrine organ; incretin (GLP-1/GIP) physiology now anchors metabolic-healthspan therapy.
2. The microbiome shapes host hormones — the estrobolome recirculates estrogen via bacterial beta-glucuronidase.
3. A failed barrier drives metabolic endotoxemia and inflammaging — the link between gut and systemic aging.

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Chapter 13 · Inflammaging & metabolic aging

The shared engine beneath endocrine decline — and the target most interventions actually hit.

Inflammaging is the chronic, sterile, low-grade inflammation that accumulates with age and predicts frailty, multimorbidity, and mortality.[1] It is fed by several hallmarks at once: cellular senescence, in which growth-arrested cells secrete a pro-inflammatory "SASP" that damages neighbors and, in mouse models, measurably shortens healthy lifespan when allowed to accumulate;[2] mitochondrial dysfunction; and dysbiosis (Chapter 12). These converge on insulin resistance — arguably the central metabolic lesion of aging.[3]

Nutrient-sensing pathways tie this together. Chronic overnutrition keeps mTOR and IGF-1 signaling high and AMPK (the fasting/energy-stress sensor) low — a pro-growth, pro-aging posture. The two most-studied pharmacologic levers act here: rapamycin inhibits mTOR and extends lifespan even when started late in mice,[4] and metformin activates AMPK and is the basis of the TAME trial's attempt to test aging itself as an indication.[5] Both remain investigational for healthy-aging use in humans.

WHY THIS CHAPTER MATTERS — Most "regenerative" interventions — exercise, time-restricted eating, GLP-1 agonists, senolytics, omega-3 — converge on the same final common path: lowering inflammatory tone and restoring insulin sensitivity. Judge any new therapy by whether it plausibly moves these needles, and whether that has been shown in humans.

CLINICAL DATA

IL-6 / CRP

elevated levels independently predict mortality, frailty, and disability in older adults.[6]

~9–14%

median lifespan extension from rapamycin in mice, even when started late.[4]

TAME

the metformin trial designed to test aging itself as an indication.[5]

CHAPTER 13 · CONTINUED

GLYCATION & AGES — THE SUGAR-AGING BRIDGE

One metabolic process ties skin, vascular, and endocrine aging together: glycation. Reducing sugars react non-enzymatically with protein amino groups to form a reversible Schiff base, then a more stable Amadori product, and finally irreversible advanced glycation end products (AGEs) — accelerated by hyperglycemia, a high-sugar or heavily processed diet, and UV exposure.[7] AGEs damage tissue two ways. Structurally, they covalently cross-link long-lived collagen and elastin, stiffening the dermis and arterial wall and producing the loss of elasticity that reads as aged skin GRADE B. Biochemically, AGEs engage their receptor RAGE, driving NF- κ B-mediated release of IL-6, TNF- α , and matrix metalloproteinases — a direct molecular link from metabolic load to inflammaging and matrix breakdown GRADE B.[7,8]

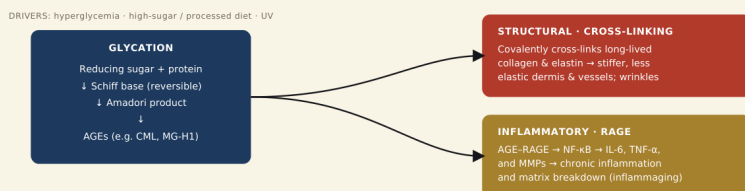


FIGURE 13.1 · THE GLYCATION CASCADE — ONE PATHWAY FROM DIETARY AND METABOLIC SUGAR LOAD TO SKIN, VASCULAR, AND METABOLIC AGING, STRUCTURALLY (CROSS-LINKING) AND INFLAMMATORILY (RAGE) · SCHEMATIC, NOT TO SCALE

The clinical levers follow the biology: glycemic control, a lower dietary-AGE pattern (less high-heat, ultra-processed food), and photoprotection. Topical and oral "anti-glycation" actives are marketed heavily but rest mostly on surrogate endpoints such as skin autofluorescence rather than hard outcomes GRADE C.[7]

These ideas rest on a now-classic literature: Franceschi's inflamm-aging hypothesis,[9] the life-course mapping of chronic inflammation onto age-related disease,[10] and Campisi's establishment of cellular senescence and its secretory phenotype as central drivers.[11]

CHAPTER 13 • CONTINUED

THREE THINGS TO REMEMBER

1. Inflammaging — chronic sterile inflammation — is a shared driver of age-related disease and frailty.
2. Senescent cells and their SASP propagate damage; clearing them extends healthspan in mice.
3. Nutrient sensing (mTOR/IGF-1 high, AMPK low) is the metabolic posture of aging; rapamycin and metformin target it but remain investigational for healthy aging.

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PART

Part III · Interventions

What to actually do – hormone optimization, peptides and bioregulators, the gut toolkit, supplements and senotherapeutics, and lifestyle and circadian medicine, each rated by evidence.

IN THIS PART

- Hormone optimization
- Peptides & bioregulators for skin and regeneration
- The gut toolkit
- Supplements & senotherapeutics
- Lifestyle & circadian optimization

Chapter 14 · Hormone optimization

Replace confirmed, symptomatic deficiency to a physiologic range — not beyond it.

The governing principle of regenerative endocrinology is restoration, not supraphysiology: treat documented, symptomatic deficiency, dose to a midnormal physiologic range for the patient's age and sex, and monitor. "More is better" is the error that gives the field its bad name.

TESTOSTERONE (MEN)

For men with symptoms plus two low morning total testosterone measurements, therapy improves sexual function, lean mass, mood, and bone density GRADE A.[1] The TRAVERSE trial found testosterone non-inferior to placebo for major adverse cardiovascular events in higher-risk men, while signaling more atrial fibrillation, venous thromboembolism, and acute kidney injury.[2] Monitor testosterone, hematocrit (hold for >54%), PSA, and symptoms; avoid if near-term fertility is desired.

ESTROGEN & PROGESTERONE (WOMEN)

For bothersome menopausal symptoms, hormone therapy is the most effective treatment, and for healthy women under 60 or within 10 years of menopause the benefits generally outweigh the risks GRADE A.[3] Transdermal estradiol carries lower thrombotic risk than oral; a progestogen is required for endometrial protection in women with a uterus. It is not approved as a primary anti-aging or cardioprotective intervention — symptom relief and bone protection are the indications.

THYROID, DHEA, AND GROWTH HORMONE

Overt hypothyroidism is treated; in older adults, subclinical hypothyroidism rarely benefits from levothyroxine, which produced no symptom improvement in the TRUST trial GRADE A — a caution against over-treating mildly elevated TSH in aging.[4] DHEA replacement is biochemically logical but disappointing in practice: a rigorous two-year trial found no benefit on body composition, physical performance, or quality of life GRADE B.[5] Growth hormone in healthy older adults is not recommended — modest body-composition changes are outweighed by edema, arthralgia, insulin resistance, and carpal tunnel, and it is illegal to prescribe for anti-aging in the US GRADE A.[6]

CHAPTER 14 • CONTINUED

THERAPY • INDICATION • CORE MONITORING

Testosterone	Symptoms + confirmed low AM total T · Total T, hematocrit, PSA, symptoms
Estrogen + progestogen	Menopausal symptoms; bone · Symptoms, bleeding, breast/CV risk
Levothyroxine	Overt hypothyroidism · TSH (avoid over-replacement in elderly)
DHEA	Documented adrenal insufficiency · DHEA-S, androgenic effects
Growth hormone	Proven GH deficiency only · IGF-1, glucose; not for aging

CLINICAL PEARL — CONFIRM BEFORE YOU REPLACE — Two morning measurements for testosterone; a clear clinical menopause picture for HT; symptoms plus an appropriately elevated TSH for thyroid. Treating a single borderline value is the commonest avoidable error in this field.

CLINICAL DATA

n≈5,246	higher-risk men in TRAVERSE: testosterone non-inferior to placebo for MACE, with signals of ↑ AF, VTE, AKI.[2]
Hct >54%	threshold to hold testosterone; also monitor PSA and symptoms.[1]
<60 yr / <10 yr	menopausal hormone therapy favours benefit when started before age 60 or within 10 years of menopause.[3]

CHAPTER 14 • CONTINUED

GLP-1 RECEPTOR AGONISTS IN AESTHETIC PRACTICE

The incretin therapies transforming metabolic medicine (Chapter 12) now reshape aesthetic practice. Rapid, large-volume weight loss strips subcutaneous fat from the face, producing the hollowed, prematurely aged look popularly called "Ozempic face" — a volume change rather than accelerated skin aging, though it unmasks pre-existing laxity; restoration (filler, fat grafting, energy-based tightening) is the usual remedy.[7] More consequential than appearance is body composition: roughly a quarter to 40% of the weight lost on a GLP-1 agonist is fat-free (lean) mass, so although the fat-to-lean ratio usually improves, absolute muscle is lost — a real concern in older or sarcopenic patients.[8] Treat a GLP-1 agonist as an endocrine intervention with aesthetic and musculoskeletal consequences: pair it with adequate protein and resistance training to defend lean mass, watch nutritional adequacy at reduced intake, and set realistic expectations about facial volume.

THREE THINGS TO REMEMBER

1. Restore to physiologic range for confirmed, symptomatic deficiency; supraphysiologic dosing is the field's signature mistake.
2. TRT is non-inferior for MACE (TRAVERSE) but watch hematocrit, AF, and VTE; HT is first-line for menopausal symptoms in appropriate women.
3. Don't treat subclinical hypothyroidism reflexively; DHEA shows no functional benefit; GH is not an anti-aging therapy.

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Chapter 15 · Peptides & bioregulators for skin and regeneration

A field rich in mechanism, thin in human outcomes, and tightly constrained by regulation.

This is where enthusiasm most outruns evidence. The honest summary: a small number of peptides have real human data for narrow indications; most "regenerative" peptides rest on preclinical and anecdotal support and, for now, have no clear lawful route to compounding in the US — a status under active FDA review in 2026.

AGENT · CLAIMED USE · EVIDENCE · STATUS

Topical GHK-Cu	Collagen, skin repair · GRADE C small trials · Cat. 1 (topical)
Tesamorelin	GHRH analog; visceral fat (HIV) · GRADE A indication · FDA-approved
Oral collagen peptide	Skin hydration/elasticity · GRADE C small RCTs · Supplement
CJC-1295 / ipamorelin	GH secretagogues, "anti-aging" · GRADE C anecdotal · Cat. 2 (not in 2026 review)
BPC-157 / TB-500	Tissue/gut repair · GRADE C preclinical · Under PCAC review, Jul 2026
Epitalon / thymalin	Bioregulator "rejuvenation" · GRADE C sparse · Epitalon under PCAC, 2026
Melanotan II	"Tanning"; libido (α MSH analog) · GRADE D harms > benefit · Unapproved / illicit

GHK-Cu, a copper-binding tripeptide, has a long mechanistic record for collagen and matrix synthesis and modest clinical support in topical use GRADE C.[1] Tesamorelin is the instructive exception — a GHRH analog that completed the regulatory path for a defined indication (HIV-associated visceral adiposity) GRADE A, showing what real peptide evidence looks like.[2] Oral collagen peptides have randomized support for skin parameters, though mostly from small, industry-funded trials GRADE C.[3]

CHAPTER 15 • CONTINUED

By contrast, the popular injectables sit a tier lower. BPC-157 and TB-500 (largely rodent data),[4] the GH secretagogues CJC-1295 and ipamorelin, and the bioregulators epitalon and thymalin[5] pair plausible mechanisms with mostly anecdotal human use rather than controlled outcome trials GRADE C.

Their regulatory status is in flux: in 2026 the FDA removed several from 503A Category 2 and referred seven peptides — BPC-157, KPV, TB-500, MOTS-c, epitalon, semax, and emideltide (DSIP) — to its Pharmacy Compounding Advisory Committee (PCAC) for possible addition to the 503A bulks list at a July 2026 meeting.[6] Three caveats matter: PCAC only advises and any change still requires FDA rulemaking, so review is not approval; CJC-1295 and ipamorelin are not in this round; and removal from Category 2 does not by itself create a lawful route to compound. Mechanism and a pending hearing are not a license to prescribe.

A NOTE ON "GLOW" AND "KLOW" BLENDS

Two marketing names recur in clinics and online. GLOW is a fixed injectable blend of GHK-Cu, BPC-157, and TB-500 — GHK-Cu-dominant and pitched for skin quality, collagen, and tissue repair. KLOW is the same three plus KPV, a Lys-Pro-Val tripeptide derived from α -MSH with anti-inflammatory activity through NF- κ B inhibition,[7] added for an immune and gut-skin angle. Users report smoother texture, fewer fine lines, clearer skin, and sometimes improved hair with topical or injectable use — but these are anecdotal and preclinical signals: there are no controlled human trials of either blend, the products are sold "research-use-only" as reconstituted powder, and the component peptides carry the 503A constraints above. Treat GLOW and KLOW as nomenclature, not evidence — the names bundle plausible mechanisms, not demonstrated outcomes.

CHAPTER 15 • CONTINUED

REAL-WORLD RUO CAUTIONS AND COMMUNITY PRACTICES

Two practical points round out the picture. First, a genuine safety signal: because research-use-only powder is made outside GMP, impurity, inaccurate dosing, and heavy-metal contamination (arsenic, lead) are documented. New hyperpigmentation, melanosis, or dark spots appearing on RUO peptides should be read as a red flag — most plausibly contamination (cutaneous hyperpigmentation is a classic sign of arsenic exposure) or overdosing of growth- or melanocortin-pathway peptides, not a benign cosmetic effect — and should prompt stopping the product and questioning its source. Second, local reactions: injectable GHK-Cu commonly causes injection-site irritation and inflammation (copper is locally irritating); standard measures help — rotate sites, lower the concentration or volume, apply a cold compress, and pause if persistent. Some users anecdotally inject KPV (an anti-inflammatory tripeptide) at the inflamed site, and others report dandruff or seborrheic scaling improving on the GLOW stack — both anecdotal, unvalidated, and not tested indications. None of these community workarounds substitutes for a regulated product and clinical oversight.

Third, a cautionary case in point. Melanotan II — an injectable α -melanocyte-stimulating-hormone (melanocortin) analog sold illicitly online as a "tanning" peptide and informal libido aid — is unapproved by the FDA and by regulators in the UK, EU, and Australia, and it concentrates the RUO hazards in one molecule. Acutely it causes nausea, facial flushing, yawning and stretching, and appetite loss, and — through central melanocortin receptors — spontaneous erections and priapism, the last a urologic emergency that can require cavernosal aspiration or surgery. Dermatologically it is the opposite of benign: by driving melanogenesis indiscriminately it darkens and enlarges existing moles, provokes eruptive and atypical (dysplastic) nevi and melanonychia, and has been linked in case reports to melanoma arising in pre-existing lesions — causation is unproven, but the mechanism (proliferation of melanocytes, including neoplastic clones) makes it a real concern, and a personal or family history of melanoma is an absolute contraindication. Serious systemic toxicity (rhabdomyolysis, neurologic effects) and the usual non-sterile-injection risks round out a profile with no place in a regenerative or aesthetic plan.[8]

CHAPTER 15 • CONTINUED

CLINICAL & MEDICOLEGAL PEARL — Before discussing any peptide, confirm its current 503A/503B status and your jurisdiction's rules. "Research-use-only" product is not a pathway to patient administration. Document the investigational nature and informed consent, and prefer agents and routes with regulatory standing.

CLINICAL DATA

~15–18%	reduction in visceral adipose tissue with tesamorelin in its approved HIV indication.[2]
No human RCTs	GLOW/KLOW blends (GHK-Cu + BPC-157 + TB-500 ± KPV) rest on anecdotal and preclinical signals only.[4]
7 peptides	referred to FDA's PCAC (Jul 2026) for possible 503A listing — review, not approval.[6]
Evidence tier	most regenerative peptides sit at grade C — mechanistic plausibility and small or anecdotal human data, not outcome trials.
Route matters	topical signaling such as GHK-Cu is better characterized than systemic injection; named "stacks" are marketing, not validated products.
Counseling frame	name the specific endpoint and its grade, keep evidence and regulatory status as separate checks, and document the investigational conversation.

THREE THINGS TO REMEMBER

1. Tesamorelin (approved, in its indication) is the model of real peptide evidence; topical GHK-Cu (C) and oral collagen (C) are reasonable.
2. BPC-157, TB-500, CJC-1295/ipamorelin, epitalon, and thymalin have plausible mechanisms but only anecdotal human data (C); seven peptides go before the FDA's PCAC in July 2026 — but review is not approval, and there is no clear lawful compounding route yet.
3. GLOW (GHK-Cu + BPC-157 + TB-500) and KLOW (+ KPV) are marketing names, not evidence; verify regulatory status and document consent before any peptide discussion.

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Chapter 16 · The gut toolkit

Most gut "regeneration" runs through diet and the barrier — not a bottle of peptides.

The highest-yield gut interventions are unglamorous and well evidenced.

Dietary fiber and minimally processed, plant-rich eating consistently improve metabolic and inflammatory markers and feed short-chain-fatty-acid-producing microbes that support the barrier GRADE A.[1]

The incretin axis — now drugged with GLP-1 and GIP/GLP-1 agonists — is the most powerful pharmacologic lever for the metabolic consequences of a dysregulated gut-endocrine system (Chapter 12).[2]

Probiotics are widely used but the evidence is strain- and indication-specific, not generic; benefits are modest and colonization is inconsistent GRADE C.[3]

Next-generation organisms such as *Akkermansia muciniphila* show early proof-of-concept for metabolic markers but remain exploratory GRADE C.[4]

The popular "leaky gut" repair stack — L-glutamine, zinc carnosine, and peptides such as BPC-157 — has mechanistic and preclinical rationale for barrier support, but human outcome data are limited, and BPC-157 carries the regulatory constraints of Chapter 15 GRADE D.[5]

SEQUENCE THAT WORKS

1. (1) Remove drivers of barrier injury — ultra-processed diet, excess alcohol, unnecessary NSAIDs/PPIs.
2. (2) Feed the microbiome with fiber and polyphenols.
3. (3) Treat metabolic disease directly (including GLP-1 agonists where indicated).
4. (4) Consider targeted probiotics for specific indications.

Reach for investigational peptides last, if at all.

CHAPTER 16 • CONTINUED

CLINICAL DATA

25–38 g/d	fiber target; each additional ~8 g/day tracks with lower all-cause mortality.[1]
Akkermansia	pasteurized <i>A. muciniphila</i> improved insulin sensitivity in a proof-of-concept RCT.[4]
Strain-specific	probiotic benefit is indication- and strain-dependent; colonization is inconsistent.[3]

THREE THINGS TO REMEMBER

1. Fiber and a plant-rich, minimally processed diet are the best-evidenced gut interventions (A).
2. Probiotics are strain- and indication-specific (C); *Akkermansia* is promising but exploratory (C).
3. Barrier-repair supplements and peptides are mechanism-rich but human-data-poor — and BPC-157 is regulated; use diet and incretins first.

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Chapter 17 · Supplements & senotherapeutics

A small number of compounds have real human trials; most do not. Grade accordingly.

The geroscience supplement market is vast and the human evidence shallow, but a handful of compounds have reached randomized human trials — the right place to focus.

Senolytics (dasatinib plus quercetin; fisetin) clear senescent cells and extend healthspan in mice;[1] first-in-human pilots in pulmonary fibrosis and diabetic kidney disease show feasibility and reduced senescent-cell burden but are small, surrogate-only, and not outcome-powered, so any human healthspan claim remains GRADE C.[2,3] Urolithin A, a mitophagy activator, is safe and improved muscle strength and endurance in randomized trials GRADE C — a muscle-function endpoint, not a proven healthspan benefit.[4,5] Spermidine induces autophagy and is cardioprotective and lifespan-extending in models, with supportive epidemiology GRADE C.[6] NAD⁺ precursors (NR, NMN) reliably raise NAD⁺ and are well tolerated, but durable outcome benefit is unproven GRADE C.[7]

Among the "boring" supplements, the evidence is paradoxically stronger for endpoints that matter. Vitamin D did not reduce cancer or cardiovascular events in the large VITAL trial, tempering enthusiasm for universal high-dose use GRADE A — correct deficiency rather than chase a longevity effect.[8] Omega-3 and adequate dietary protein support muscle and metabolic health and belong in most regenerative plans.

The clinical posture follows from the evidence. Build the foundation from what has human outcome data — protein, omega-3, corrected vitamin D, and the lifestyle levers of the next chapter — then add at most one or two mechanistically plausible agents tied to a defined endpoint, such as urolithin A for muscle endurance, rather than stacking a dozen unproven compounds. Each addition should earn its place: a reason to start, a target to watch, a trial period, and an honest review of whether anything measurably changed. Most of the market fails that test, and the opportunity cost is real — cost, pill burden, and the false reassurance that displaces interventions that actually work.

CHAPTER 17 • CONTINUED

COMPOUND • BEST HUMAN SIGNAL • GRADE

Urolithin A	Muscle strength/endurance (RCT) • GRADE C
Senolytics (D+Q, fisetin)	Human healthspan (pilots, surrogate only) • GRADE C
Spermidine	Autophagy; cohort associations • GRADE C
NAD ⁺ precursors (NR/NMN)	Raises NAD ⁺ ; outcomes unproven • GRADE C
Vitamin D	Correct deficiency (not prevention) • GRADE A

CLINICAL PEARL — Rank by trial quality, not by marketing. Urolithin A and corrected vitamin-D deficiency have the firmest footing; NAD⁺ precursors, spermidine, and senolytics are reasonable-to-discuss but investigational. Senolytics in particular should generally be confined to clinical trials.

CLINICAL DATA

500 mg/d	urolithin A dose improving muscle endurance and mitophagy markers in RCTs.[4]
n=25,871 (VITAL trial)	vitamin D did not cut cancer or CVD; correct deficiency, do not chase prevention.[8]
Pilot-stage	senolytic dasatinib+quercetin reduced senescent-cell burden in small IPF and diabetic-kidney studies.[2]

Human translation remains early. The proposed TAME trial — metformin tested against composite aging endpoints — has been repeatedly delayed,[9] and first-in-human senolytic studies (dasatinib plus quercetin) clear senescent cells but show inconsistent effects on clinical and epigenetic-clock endpoints.[3,10] The field's momentum — from geroscience consortia to the \$101M XPRIZE Healthspan competition — has not yet produced an approved gerotherapeutic.

THREE THINGS TO REMEMBER

1. Focus on compounds with randomized human data: urolithin A (C; a muscle-endurance endpoint) leads the geroscience supplements.
2. NAD⁺ precursors raise NAD⁺ but lack outcome data; spermidine is promising-but-investigational (C) and senolytics remain investigational with no human outcome data (D).
3. Vitamin D: correct deficiency rather than supplement universally (VITAL was null for cancer/CVD).

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CHAPTER 17 • CONTINUED

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Chapter 18 · Lifestyle & circadian optimization

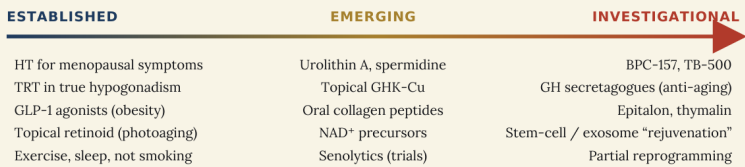
The interventions with the strongest evidence cost the least and carry no Category-2 label.

If regenerative endocrinology has a foundation, it is here. The lifestyle levers outperform every supplement and most drugs for healthspan, and they amplify whatever else is added.

Exercise is the closest thing to a polypill: a combination of resistance training (to preserve muscle, bone, and insulin sensitivity) and aerobic conditioning improves nearly every hallmark of aging GRADE A.[1]

Nutrition matters most as a pattern: a Mediterranean-style, minimally processed, protein-adequate diet has randomized cardiovascular-prevention evidence GRADE A,[2] and meal timing aligned to the active phase improves insulin sensitivity independent of weight loss.[3]

Sleep is non-negotiable endocrine medicine: both short and long sleep associate with higher mortality, and most GH secretion and tissue repair are sleep-dependent GRADE B.[4] Protecting it starts with light hygiene — bright morning light, dim evening light — given how readily evening light suppresses melatonin.[5] Chronic psychological stress, via sustained cortisol, opposes all of the above; stress regulation is a legitimate endocrine intervention.



Position reflects strength of human evidence and regulatory status, not promise

FIGURE 18.1 · WHERE EACH INTERVENTION SITS ON THE EVIDENCE-TO-INVESTIGATIONAL SPECTRUM · SCHEMATIC, NOT TO SCALE

CHAPTER 18 • CONTINUED

CLINICAL DATA

Polypill	resistance + aerobic training improves nearly every hallmark of aging (grade A).[1]
~30%	relative reduction in major cardiovascular events on a Mediterranean diet (PREDIMED).[2]
1.0–1.2 g/kg/d	protein target for older adults to preserve muscle, paired with resistance training.[6]
7–9 h	sleep associated with lowest mortality; both short and long sleep raise risk.[4]

THREE THINGS TO REMEMBER

1. Resistance + aerobic exercise is the highest-yield regenerative intervention (A) and amplifies everything else.
2. A Mediterranean-style pattern (A) plus active-phase meal timing beats any single supplement.
3. Sleep and light hygiene are endocrine therapy: protect the sleep window; get morning light, avoid evening light.

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PART

Part IV · Frontier & Practice

The regenerative frontier and how to put it together – cell and tissue replacement, and a synthesis for practice.

IN THIS PART

- Regenerative therapies for endocrine organ & tissue replacement
- Putting it together

Chapter 19 · Regenerative therapies for endocrine organ & tissue replacement

The horizon of the field: rebuilding the gland itself rather than replacing its output.

Everything to this point has aimed at optimizing tissue that still functions. Regenerative endocrinology in its most ambitious sense aims further — to repair, replace, or regenerate the endocrine tissue itself and restore native, regulated hormone production rather than approximating it with exogenous replacement.[1,2] A transplanted, glucose-responsive beta-cell mass is a categorically different thing from injected insulin; the promise is closed-loop physiology, not open-loop dosing. Most of this work is investigational, but one application has now reached early-phase (phase 1–2) human trials, so the category can no longer be dismissed as speculative.[3]

THE TOOLKIT

Induced pluripotent stem cells (iPSCs) are adult somatic cells — a skin or blood sample — reprogrammed to an embryonic-like state with defined factors,[4] then differentiated toward a target endocrine lineage.[5] Mesenchymal stromal cells (MSCs) are adult cells used less to become the missing tissue than to support it: they secrete growth factors, dampen inflammation, and promote endogenous repair — a paracrine, not a replacement, strategy GRADE C.[6] Tissue engineering combines cells with bioengineered scaffolds to build implantable, vascularized constructs that could one day function as transplantable glands.[7]



Most clinical translation still requires immunosuppression or immune-evasion engineering

FIGURE 19.1 · THE SHARED PIPELINE: SOURCE CELL → REPROGRAM/EXPAND → DIFFERENTIATE → DELIVER · SCHEMATIC

CHAPTER 19 • CONTINUED

CLINICAL PEARL • TESTING — When a patient arrives citing a "stem-cell" clinic for a hormone problem, two questions clarify almost everything: what cell type, delivered how, and is there a registered trial? Autologous MSC infusions marketed for "endocrine rejuvenation" have essentially no controlled outcome data; legitimate cell-replacement work is concentrated in registered trials with defined products and endpoints.

WHERE EACH ORGAN STANDS**TARGET • LEADING APPROACH • STATUS**

Pancreas (T1D)	iPSC/ESC-derived islet (beta) cells; encapsulation to reduce immunosuppression • GRADE B
Thyroid	iPSC → functional follicular cells (groundwork for biologic transplant) • GRADE D
Ovary / endometrium	Platelet-rich plasma and stem-cell approaches for endocrine/regenerative repair • GRADE D
Adrenal / pituitary	Stimulating resident progenitor cells to regenerate steroidogenic tissue • GRADE D

Diabetes leads decisively. Stem-cell-derived islet products have moved from concept to early-phase clinical data: in 2023 the FDA approved the first cellular therapy for type 1 diabetes (donislecel, a cadaveric-islet product),[8] and in 2025 a stem-cell-derived islet therapy (zimislecel) reported that all treated patients recovered endogenous insulin secretion, with most insulin-independent at one year — albeit on chronic immunosuppression.[9] Thyroid work has coaxed pluripotent stem cells into functional follicular cells, groundwork for biologic transplants.[5] In reproductive endocrinology, platelet-rich plasma and stem-cell approaches are in early, mostly uncontrolled trials for ovarian and endometrial regeneration.[10] For the adrenal and pituitary, research is still at the level of stimulating resident progenitor cells.[11]

CLINICAL PEARL • THERAPY — The recurring catch is immune tolerance. Even a perfectly differentiated islet graft is foreign tissue; the durable wins will come from encapsulation or hypoimmune engineering that frees patients from lifelong immunosuppression — not from the differentiation protocol alone. Counsel patients that "regenerated" does not yet mean "immunosuppression-free."

CHAPTER 19 • CONTINUED

The honest summary: immense promise, narrow delivery. Most regenerative endocrine therapies remain in research or trial phases, and translation faces scientific and policy barriers alike.[12] For current status, the ISSCR's guidelines for stem-cell research and clinical translation and ClinicalTrials.gov are the right references[13] — and the right thing to give a patient is a trial registry link, not a clinic brochure.

For the practitioner, the near-term reality is narrower than the headlines. The first durable wins are arriving where the target is a single, well-defined cell type with a clear functional readout — pancreatic beta cells for type 1 diabetes — while multicellular organs with intricate architecture and feedback loops remain far from the clinic. That gap is precisely where unregulated "stem-cell" and "regenerative" clinics market unproven infusions for everything from anti-aging to hormonal decline, often at high cost and real risk. The defensible stance is to follow the registered trials, use approved products only within their narrow indications, and steer patients away from cash-pay clinics selling a regeneration the evidence has not yet delivered.

CLINICAL DATA

FORWARD, 2025	all treated recovered endogenous insulin secretion; most insulin-independent at 1 yr (on immunosuppression).[9]
2023	donislecel (Lantidra) — first FDA-approved cellular therapy for type 1 diabetes.[8]
OSKM, 2006	Yamanaka factors reprogram somatic cells to iPSCs — the engine of cell-replacement therapy.[4]

THREE THINGS TO REMEMBER

1. Goal: rebuild regulated, native hormone production — not replace output. Tools are iPSCs, MSCs (paracrine), and tissue engineering.
2. Diabetes leads: an FDA-approved cellular therapy (2023) and early-phase (phase 1-2) stem-cell-derived islet data (zimislecel, 2025); thyroid, ovarian, and adrenal work remain preclinical-to-early.
3. Immune tolerance, not differentiation, is the rate-limiting step. Most therapies are investigational; point patients to registered trials, not unregulated clinics.

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CHAPTER 19 • CONTINUED

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Chapter 20 · Putting it together

A responsible framework for the regenerative-endocrinology consult.

The patient who comes in asking about peptides, hormones, and "regeneration" is asking a reasonable question inside a field crowded with hype. The clinician's job is to separate the levers that work from the ones that merely sell — and to sequence them honestly.

AS AN ORDER OF OPERATIONS

1. Build the foundation first. Exercise, nutrition, sleep, and circadian alignment are the only interventions with grade-A healthspan evidence, and they amplify everything layered on top. No peptide rescues a patient who does not sleep.
2. Correct what is measurably deficient. Replace hormones to a physiologic target when there is a genuine deficiency and a guideline to support it — testosterone in diagnosed hypogonadism,[1] menopausal hormone therapy where indicated and not contraindicated.[2]
3. Add evidence-graded adjuncts. A short list — urolithin A, corrected vitamin D, GLP-1 agonists where indicated — has real human data.
4. Treat everything else as investigational.

CLINICAL PEARL · THE REGULATORY LINE — Many popular regenerative peptides — BPC-157, TB-500, CJC-1295, ipamorelin, epitagon among them — were placed by the FDA into the 503A Category-2 list, leaving no clear lawful pathway to compound them for human use; as of mid-2026 this is under active reconsideration (a PCAC meeting in July 2026 is weighing seven docketed peptides — BPC-157, KPV, TB-500, MOTS-c, DSIP, semax, and epitagon — for the bulks list; CJC-1295 and ipamorelin are not in that round).[3] "Research use only" is not a clinical pathway. Document that the patient understands a compound's regulatory status before any prescription, and do not present unapproved peptides as established protocols.

WHAT TO PROMISE, AND WHAT NOT TO

Promise rigor, not reversal. The defensible message is that aging is modifiable at the margins by well-evidenced means, that hormone optimization is replacement medicine with rules, and that the frontier — cell replacement, partial reprogramming — is genuinely exciting and genuinely not ready.[4] A clinician who holds that line builds the one thing this field is short on: trust.

CHAPTER 20 • CONTINUED

CLINICAL DATA

Grade A	lifestyle (exercise, diet, sleep), guideline-based HT/TRT, and GLP-1 agonists where indicated.[1]
Adjuncts	urolithin A (C, muscle) and correcting vitamin-D deficiency have the firmest data; NAD ⁺ , spermidine, senolytics remain investigational.
503A flux	peptides have moved on, off, and into review on the bulk-substance lists; seven face PCAC in Jul 2026. Review is not approval — no settled lawful compounding route yet.[3]

THREE THINGS TO REMEMBER

1. Sequence interventions: foundation (lifestyle, A) → correct true deficiency (guideline-based HT/TRT) → evidence-graded adjuncts → clearly-labeled investigational.
2. Know and disclose regulatory status: the FDA 503A peptide lists are unsettled — substances have moved on, off, and into PCAC review in 2026, and review is not approval, so no settled lawful compounding route exists yet. Verify before prescribing.
3. Promise rigor, not rejuvenation. The frontier is real but not ready; honest framing is the foundation of trust.

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The guide at a glance

Five levers, one principle: act on what is established, discuss what is emerging, and flag what is investigational – and let the evidence grade attach to a specific use, never to a global “anti-aging” promise.

THE THROUGHLINE

The biggest gains are unglamorous and well evidenced; the most marketed interventions are the least proven. When the two conflict, follow the evidence, name the regulatory status, and keep the patient’s healthspan – not the trend – at the center.

HORMONES

Replace true deficiency to guideline targets – TRT, menopausal HT, thyroid – and use GLP-1/GIP agonists for obesity. The strongest, most actionable lever.

EVIDENCE — GRADE

Hormones	GRADE A
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GUT

Feed the barrier and microbiome; incretin (GLP-1/GIP) biology ties gut health to metabolic healthspan. Diet and fiber come first.

EVIDENCE — GRADE

Gut	GRADE A
Gut	GRADE C

LIFESTYLE & CIRCADIAN

Exercise, nutrition, sleep, and light timing are the highest-evidence, lowest-cost interventions in the book. Everything else is adjunctive.

EVIDENCE — GRADE

Lifestyle & circadian	GRADE A
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PEPTIDES & BIOREGULATORS

Mechanistically interesting but clinically thin; most are investigational and in regulatory flux. Topical GHK-Cu is the modest exception.

EVIDENCE — GRADE

Peptides & bioregulators	GRADE C
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THE GUIDE AT A GLANCE • CONTINUED

REGENERATION

Cell and tissue replacement is real in diabetes (islet therapy); elsewhere it is preclinical-to-early. Avoid unregulated “stem-cell” clinics.

EVIDENCE — GRADE

Regeneration	GRADE A
Regeneration	GRADE D

Clinical pearls

PEARL 1 — Foundation before frontier. Exercise, nutrition, sleep, and circadian alignment are the only grade-A healthspan levers and they amplify everything else. No peptide compensates for a patient who does not sleep or train.

PEARL 2 — Replace deficiency, do not chase youth. Hormone therapy is replacement medicine with rules — testosterone for diagnosed hypogonadism, MHT where indicated. Treat to a physiologic target against a deficiency, not to a "younger" number.

PEARL 3 — Evidence grade and legal status are different axes. A compound can be popular, plausible, and still neither approved nor lawfully compounded. Check both before prescribing, and document the conversation.

PEARL 4 — Most peptides are grade C. BPC-157, TB-500, CJC-1295, ipamorelin, epitalon and similar agents have thin human data, and their FDA 503A status is unsettled: the seven docketed peptides (BPC-157, KPV, TB-500, MOTS-c, DSIP, semax, epitalon) are under PCAC review in 2026 — CJC-1295 and ipamorelin are not in that round — and review is not approval. Present them as investigational, never as protocols.

PEARL 5 — The skin is an endocrine organ. It synthesizes and responds to steroids, vitamin D, and neuropeptides. Estrogen loss drives much of the collagen and hydration decline of menopausal skin — a useful counseling frame.

PEARL 6 — The gut sets the endocrine tone. The microbiome regulates GLP-1, the estrobolome, and systemic inflammation. Fiber and food-pattern change outperform most probiotic products, whose effects are strain-specific.

CLINICAL PEARLS • CONTINUED

PEARL 7 — Inflammaging is the common pathway. Chronic low-grade inflammation links the hallmarks of aging. What lowers it — exercise, metabolic health, sleep — beats any single anti-inflammatory supplement.

PEARL 8 — GLP-1 agonists earned their grade A. Semaglutide and tirzepatide have randomized weight and cardiometabolic outcome data — the strongest "regenerative-adjacent" pharmacology here. Indication and monitoring still apply.

PEARL 9 — Urolithin A leads the geroscience supplements. It has randomized human data for muscle endurance and mitophagy (grade C). Most others raise biomarkers without proven outcomes.

PEARL 10 — Senolytics belong in trials. The preclinical story is compelling; human outcome evidence is not yet there. Intermittent dasatinib + quercetin and similar regimens should generally stay within registered studies.

PEARL 11 — Cell replacement is real — in diabetes, for now. An FDA-approved cellular therapy (2023) and early-phase (phase 1–2) stem-cell-derived islet data (zimislecel, 2025) exist; thyroid, ovarian, and adrenal work remain early. Immune tolerance, not differentiation, is the limiting step.

PEARL 12 — Send patients to registries, not clinics. For any "stem-cell" inquiry, the right artifact is a ClinicalTrials.gov search and the ISSCR's stem-cell guidance — not a cash-pay clinic brochure.

PEARL 13 — Protect lean mass through any weight loss. A quarter to 40% of the weight lost on a GLP-1 agonist can be fat-free mass, and rapid loss also depletes facial and dermal fat. Pace the titration, secure adequate protein with resistance training, and raise the aesthetic trade-offs before starting — most pointed in older or sarcopenic patients.

CLINICAL PEARLS • CONTINUED

PEARL 14 — Cortisol is a rhythm, not a number. Map a flattened, evening-heavy curve with four-point salivary testing and reset it with morning light, a protected sleep window, and exercise. Reserve formal hypercortisolism work-up and referral for genuine red flags — purple striae, proximal myopathy, easy bruising — not for the lifestyle-and-fluid "cortisol face."

Testing & interpretation reference

A BASELINE PANEL FOR THE HEALTHSPAN CONSULT

DOMAIN • REPRESENTATIVE TESTS

Gonadal axis (men)	Total & free testosterone (AM, fasting, ×2), SHBG, LH, FSH, prolactin; iron studies (iron, TIBC, ferritin); semen analysis if fertility a concern
Gonadal axis (women)	Estradiol, FSH, LH, AMH (ovarian reserve), day-21 progesterone (confirms ovulation); symptoms > numbers in menopause
Hyperandrogenism	Total & free testosterone, androstenedione, DHEA-S, 17-OH-progesterone, prolactin, LH/FSH, estradiol, progesterone, IGF-1, HbA1c; 1 mg overnight DST if clinically indicated
Thyroid	Full panel: TSH, free T4, total T3, anti-TPO antibodies
Somatotropic axis	IGF-1 (interpret by age/sex; not a fountain-of-youth test)
Adrenal	Morning cortisol to exclude insufficiency (ACTH-stimulation test to confirm); 1 mg overnight dexamethasone-suppression test to screen hypercortisolism; ± DHEA-S
Metabolic	Fasting glucose, HbA1c, fasting insulin/HOMA-IR, lipids
Inflammation / status	hs-CRP, 25-OH vitamin D, CBC, CMP

TESTING & INTERPRETATION REFERENCE • CONTINUED

READING THE PANEL — CHECKLIST

- Confirm before treating: repeat an abnormal androgen or thyroid value before acting; single values mislead.
- Timing and fasting matter: testosterone and cortisol are diurnal; draw AM. Insulin and lipids fasting.
- Reference by age and sex: IGF-1 and DHEA-S decline normally with age — "low-for-young" is often normal-for-age.
- Treat the patient, not the number: a deficiency plus symptoms plus a guideline supports therapy; an isolated number does not.
- Screen for non-classic CAH in cutaneous hyperandrogenism: non-classic 21-hydroxylase deficiency can present first in the skin — treatment-resistant acne, hirsutism, or premature/exaggerated adrenarche — with an elevated morning 17-OH-progesterone (ACTH-stimulated if borderline) the key screen.
- Use a reliable assay: prefer LC-MS/MS for sex steroids and 17-OH-progesterone; older direct immunoassays are unreliable in the low (female, pediatric) range. Measure free testosterone by equilibrium dialysis or calculate it from total testosterone and SHBG.
- Time female draws to the cycle: sample androgens and 17-OH-progesterone in the early follicular phase (days 2–5); a day-21 (mid-luteal) progesterone confirms that ovulation occurred.
- Confirm a positive suppression test: oral estrogen and combined contraceptives raise cortisol-binding globulin and cause false-positive 1 mg dexamethasone tests, as do poor absorption and rapid metabolism — confirm hypercortisolism with a second modality before referral.
- Re-derive risk, not youth: the goal of the panel is cardiometabolic and endocrine risk, not a "biological age" score.

TWO-LINE RULE OF THUMB

TWO-LINE RULE OF THUMB — For diagnosis and replacement, standard endocrine testing is mature and decision-changing. For "optimization" beyond physiologic targets, the evidence thins quickly — treat such requests as a conversation about risk, not a lab to be normalized upward.

Drug safety, monitoring & contraindications

A non-exhaustive bedside reminder of the principal cautions and monitoring that accompany the therapies in this guide. It does not replace full prescribing information or current specialty guidelines — confirm contraindications, baseline workup, and monitoring intervals against those sources for each patient.

HORMONE THERAPIES

THERAPY · KEY CONTRAINDICATIONS & CAUTIONS · BASELINE &

DRUG SAFETY, MONITORING & CONTRAINDICATIONS • CONTINUED**MONITORING • STOP/REFER**

Testosterone (TRT)	Active prostate or breast cancer; erythrocytosis; untreated severe OSA; desire for near-term fertility (suppresses spermatogenesis); recent MI/stroke or uncontrolled heart failure. · Confirm low testosterone on two morning samples; baseline hematocrit, PSA (age-appropriate), lipids. Recheck testosterone, hematocrit, PSA at ~3, 6, 12 months, then annually. · Hematocrit > ~54%, or new/rising PSA or prostate nodule → hold and refer urology.
Menopausal HT	History of breast or estrogen-sensitive cancer; unexplained vaginal bleeding; active/prior VTE or known thrombophilia; active liver disease; prior stroke/CHD. Oral estrogen carries higher VTE/stroke risk than transdermal; add a progestogen if the uterus is intact. · Blood pressure, individualized VTE/breast-risk assessment, age-appropriate mammography. Use the lowest effective dose; reassess benefit–risk and duration at least annually. · New VTE, breast lump, or unexplained bleeding → stop and evaluate.
GLP-1 / GIP agonists	Personal/family history of medullary thyroid carcinoma or MEN-2 (boxed warning); prior pancreatitis; gallbladder disease; severe gastroparesis; pregnancy. Hold perioperatively per anesthesia guidance (aspiration risk). · Weight and GI tolerance; protect lean mass with resistance training and adequate protein as weight falls; watch hydration/renal function during GI losses. · Severe persistent abdominal pain (pancreatitis/gallbladder) or neck mass → evaluate.
Thyroid hormone	Avoid over-treatment: a suppressed TSH raises the risk of atrial fibrillation and bone loss, especially in postmenopausal women and the elderly. Start low and titrate in older or cardiac patients. · TSH ($\pm$ free T4) ~6–8 weeks after any dose change, then periodically; target an age-appropriate TSH and avoid suppression unless specifically indicated. · Palpitations/AF or persistently low TSH → reduce dose; cardiac symptoms → evaluate.

DRUG SAFETY, MONITORING & CONTRAINDICATIONS • CONTINUED

GH / IGF-1	For confirmed deficiency only — not for healthy aging. Contraindicated in active malignancy. Adverse effects: fluid retention/edema, arthralgia, carpal tunnel, insulin resistance and hyperglycemia. · IGF-1 titrated to an age-appropriate (not supraphysiologic) level; glucose/HbA1c. Manage only within specialist care. · Do not use for anti-aging; refer to endocrinology for true deficiency.
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MONITORING SCHEDULE — TESTOSTERONE, MENOPAUSAL HT, GLP-1**AGENT • BEFORE STARTING • FOLLOW-UP • WATCH / ACT ON**

Testosterone	Total testosterone ×2 (AM, fasting); hematocrit; PSA (age-appropriate) ± DRE; lipids. Discuss fertility goals (TRT suppresses spermatogenesis) and screen for obstructive sleep apnea. · Testosterone, hematocrit, and PSA at ~3, 6, and 12 months, then annually. · Hematocrit > ~54% → hold/reduce; rising PSA or new nodule → urology; worsening OSA.
Menopausal HT	Blood pressure; personal/family VTE and breast-cancer risk; age-appropriate mammography. Prefer transdermal estrogen when VTE risk is a concern (lower risk than oral); add a progestogen if the uterus is intact. · Reassess symptoms, dose, and benefit–risk at least annually; mammography per screening schedule. · New VTE, breast lump, or unscheduled vaginal bleeding → evaluate.
GLP-1 / GIP	Personal/family history of MTC or MEN-2; prior pancreatitis or gallbladder disease; baseline weight. Counsel on nausea and slow titration. · GI tolerance and weight at each titration; protect lean mass with resistance training and adequate protein; watch hydration/renal function during GI losses. · Severe persistent abdominal pain (pancreatitis/gallstones), neck mass, or rapid lean-mass loss → evaluate.

GEROSCIENCE & INVESTIGATIONAL AGENTS

DRUG SAFETY, MONITORING & CONTRAINDICATIONS • CONTINUED**AGENT • CAUTIONS & STATUS • MONITORING**

Senolytics (dasatinib + quercetin; fisetin) GRADE C	Confine to clinical trials. Dasatinib carries cytopenias, fluid retention/effusions, QT and bleeding risk; no validated outpatient regimen for aging. · No validated monitoring outside a trial protocol.
NAD ⁺ precursors, spermidine, urolithin A	Generally well tolerated; outcome benefit is limited (urolithin A GRADE C for muscle endurance; others GRADE C or lower). Not substitutes for established care. · Clinical follow-up; no specific labs required.
Rapamycin / metformin for longevity	Investigational for healthy aging (rapamycin GRADE C for longevity; metformin GRADE C, TAME pending). Use only within approved indications or trials. · Per the approved indication only.
Investigational peptides (BPC-157, TB-500, CJC-1295/ipamorelin, epitalon, thymalin) GRADE C	No validated human monitoring. Regulatory status unsettled (503A Category-2; of these, none of CJC-1295/ipamorelin are in the 2026 PCAC round — that round covers seven other docketed peptides; review is not approval). Immunogenicity and impurity/characterization concerns; document investigational status and informed consent. · No validated monitoring; verify current regulatory status (right).

REGULATORY STATUS — VERIFY BEFORE PRESCRIBING — Peptide compounding status changes quickly. Scan the code (or visit the FDA Human Drug Compounding pages) for the current 503A/503B Category-2 list and Pharmacy Compounding Advisory Committee actions before relying on any status quoted in this guide. Confirm your state board's rules as well.

Aligned with Endocrine Society and The Menopause Society guidance, American Thyroid Association recommendations, and current FDA labeling; this table summarizes and does not replace them. Doses are deliberately omitted — consult primary prescribing information.

The regenerative endocrinology consult

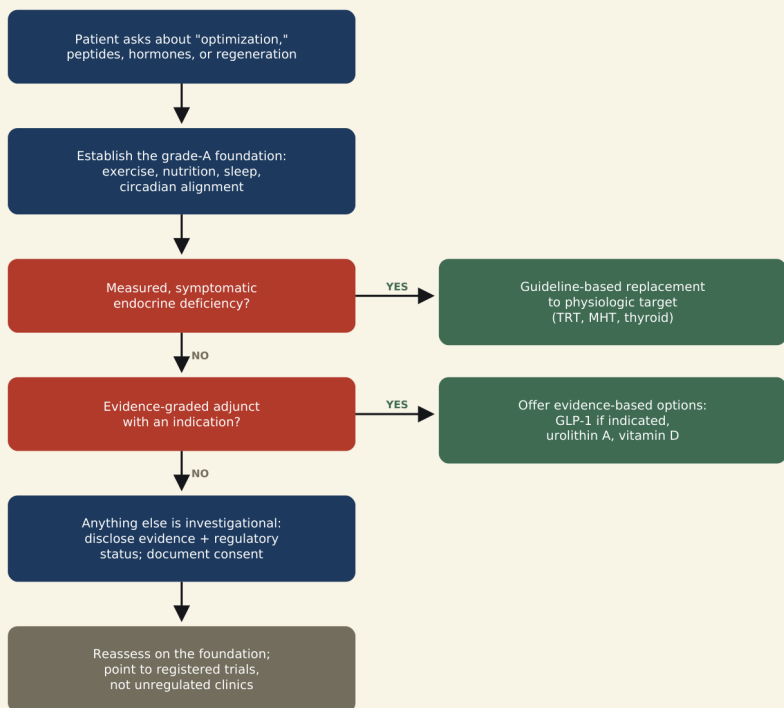
A conservative route for the patient who arrives asking about peptides, hormones, and "regeneration." The order is deliberate: the validated levers come first, replacement is reserved for genuine deficiency, and everything else is named as investigational under informed consent.

SEQUENCING THE HEALTHSPAN / REGENERATIVE CONSULT

PRINCIPLES

- Foundation first, regardless of any request.
- Replace deficiency to a physiologic target, not to "youth."
- Evidence grade and regulatory status are separate checks; clear both.
- Investigational means trials and consent, not protocols.

THE REGENERATIVE ENDOCRINOLOGY CONSULT · CONTINUED



SEQUENCING THE HEALTHSPAN / REGENERATIVE CONSULT

PRINCIPLES

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Regenerative therapy: referral pathway

A practical route for the increasingly common "should I try stem cells?" question. Diabetes is the one endocrine indication with approved and pivotal-stage cellular therapy; for everything else, the honest deliverable is a trial registry, not a clinic.

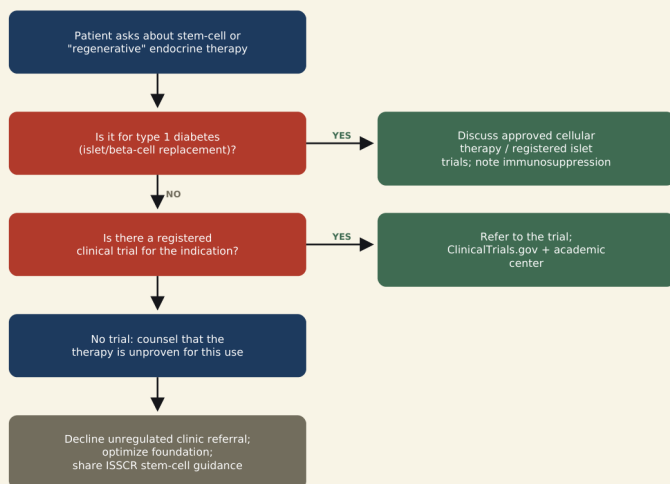
TRIAGING A REGENERATIVE / STEM-CELL REQUEST

COUNSELING POINTS — "Regenerated" does not mean "immunosuppression-free."

- "Regenerated" does not mean "immunosuppression-free."
- "Research use only" is not a clinical pathway.
- Autologous MSC infusions for "endocrine rejuvenation" have essentially no controlled outcome data.
- The right artifact for a patient is a registry link.

A NOTE ON STATE STEM-CELL LAWS — A few states now permit non-FDA-approved cell therapies in narrow cases. Florida's SB 1768 (effective July 1, 2025) lets a Florida-licensed MD or DO administer stem-cell therapies the FDA has not approved — but only for orthopedic, wound-care, and pain-management indications, with cGMP-compliant, ethically sourced cells, informed consent, and a "not FDA-approved" advertising disclosure (Texas (HB 810) and Utah have related laws). Crucially, federal law still governs, and none of these statutes covers the endocrine, aesthetic, or longevity uses in this guide — "it's legal in Florida" does not extend to a regenerative-endocrine or cosmetic indication.

REGENERATIVE THERAPY: REFERRAL PATHWAY • CONTINUED



TRIAGING A REGENERATIVE / STEM-CELL REQUEST

COUNSELING POINTS

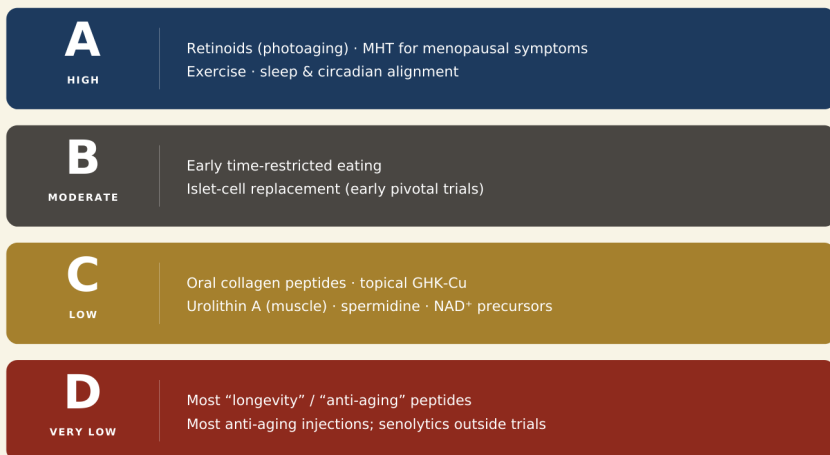
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Clinical evidence hierarchy

A single ladder for the bedside: where the interventions in this guide actually sit, strongest evidence at the top. Read it as a triage tool — act on the top rungs, discuss the middle, and present the bottom as investigational. The representative items are illustrative, not exhaustive.

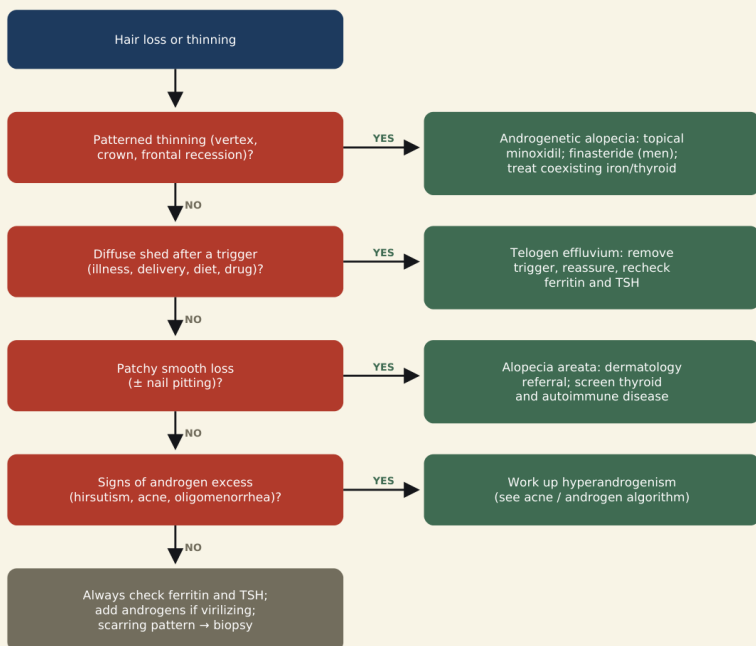


STRENGTH-OF-EVIDENCE LADDER FOR THE INTERVENTIONS IN THIS GUIDE · SCHEMATIC, NOT TO SCALE

How to read it. Evidence grades reflect evidence quality for the *specific indication discussed*, not biological plausibility or mechanistic elegance. A grade attaches to a named outcome — collagen peptides earn a C for skin hydration, not a global “anti-aging” claim — and **regulatory status is a separate axis** (a compound can be plausible, popular, and still neither approved nor lawfully compounded; see Chapter 3).

Algorithm — hair loss workup

A first-pass framework for the patient presenting with hair loss. The aim is to separate the common patterns quickly and to catch the endocrine and nutritional contributors that change management.

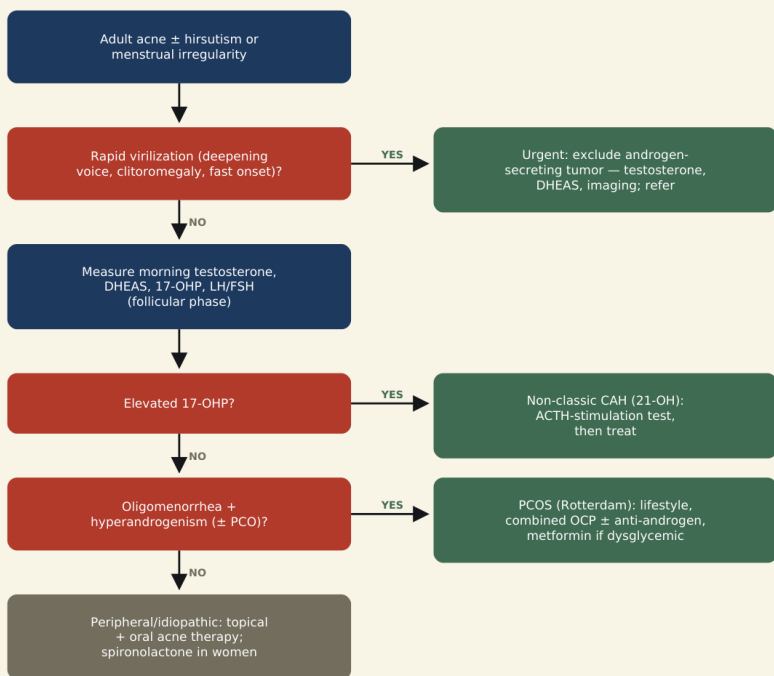


HAIR LOSS — FIRST-PASS ENDOCRINE AND DERMATOLOGIC WORKUP

Algorithm — adult acne & androgen excess

Adult acne — especially persistent, jawline, or hormonally patterned — warrants a brief endocrine screen. Rapid virilization is the one finding that demands an urgent work-up.

urgent work-up.

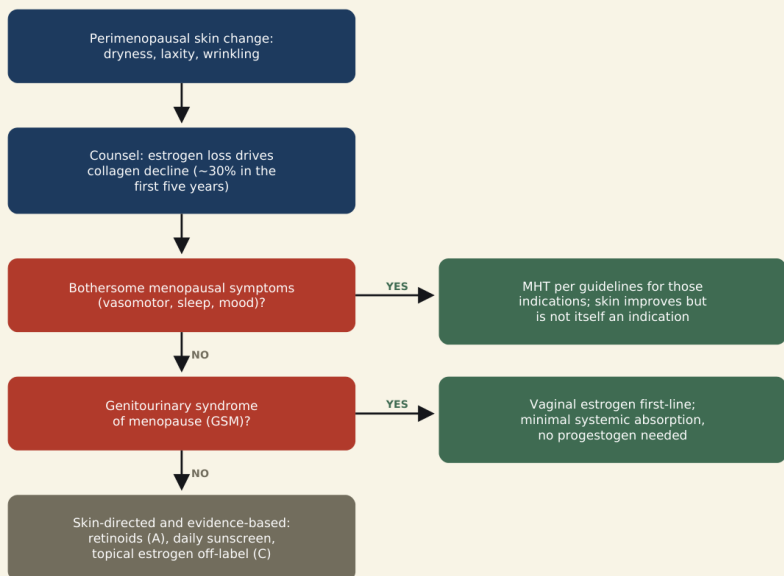


ADULT ACNE WITH SUSPECTED ANDROGEN EXCESS

Algorithm — menopausal skin aging

How to think about menopausal skin change without over-prescribing.

Systemic hormone therapy is justified by menopausal symptoms, not by the skin; skin benefit is real but secondary.

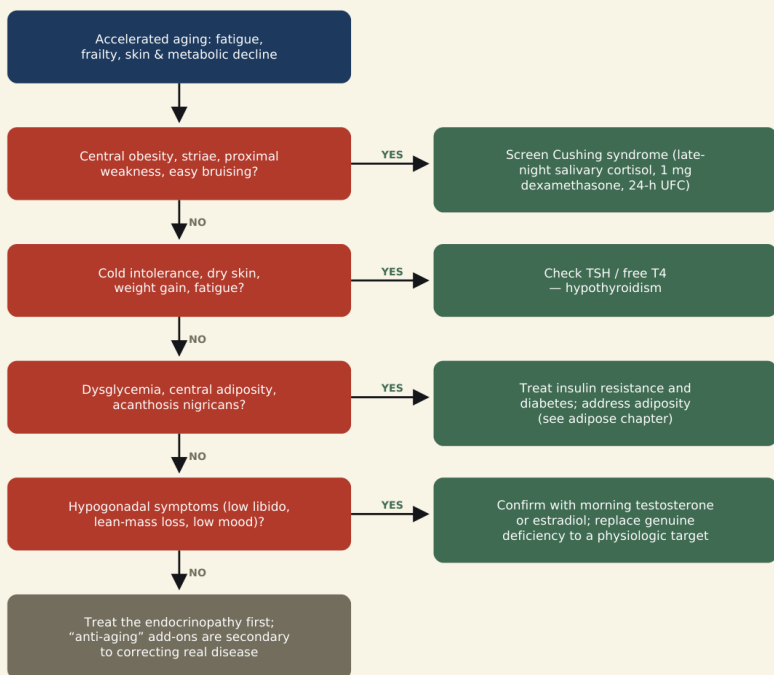


MENOPAUSAL SKIN AGING — A CONSERVATIVE PATHWAY

Algorithm — endocrine causes of accelerated aging

Before reaching for "anti-aging" tools, exclude the treatable endocrinopathies that mimic accelerated aging. Correcting real disease outperforms any optimization add-on.

optimization add-on.

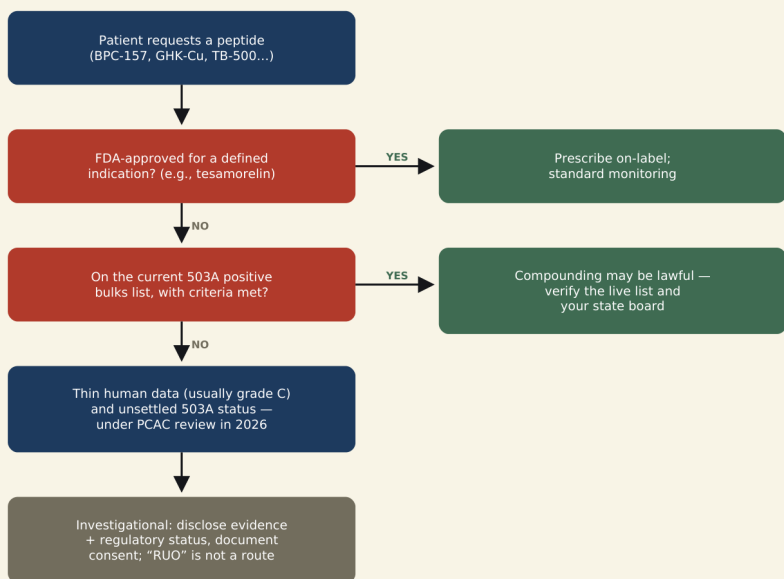


SCREENING FOR ENDOCRINE DRIVERS OF ACCELERATED AGING

Algorithm — peptide decision framework

A conservative decision path for peptide requests, mirroring Chapter 3: approval first, then a lawful compounding route, and otherwise an honest investigational conversation.

DECIDING WHETHER TO USE A REQUESTED PEPTIDE



Inherited endocrine syndromes & the skin

Cutaneous clues to genetic endocrinopathies — and the oncogene and cyclic-AMP pathways behind them.

Many familial endocrinopathies announce themselves on the skin before the gland declares itself, and a few signs are close to pathognomonic. Reading them turns a cosmetic complaint into an early diagnosis — and a referral.[1] The mechanisms are not random: most cluster around two molecular themes that this chapter uses as its organizing logic.

TWO MOLECULAR THEMES

The first is the tumor-suppressor / oncogene axis. Loss of a suppressor (menin in MEN1, neurofibromin in NF1, PTEN in Cowden) or activation of an oncogene (the RET receptor tyrosine kinase in MEN2) drives the benign skin tumors — angiofibromas, neurofibromas, trichilemmomas — that flag the syndrome. The second is the cyclic-AMP–protein-kinase-A (PKA) axis. Because cAMP/PKA signaling governs both melanocyte and endocrine-cell behavior, variants that disinhibit it produce a characteristic pairing of pigmented skin lesions with hormone excess. These two themes recur below.

MULTIPLE ENDOCRINE NEOPLASIA

MEN1 (Wermer syndrome) follows loss of the tumor suppressor menin. Its cutaneous signature is striking and underused diagnostically: multiple facial angiofibromas (~88% of patients), truncal collagenomas (~72%), and lipomas, often years before the parathyroid, pituitary, or pancreatic tumors are found.[2] These angiofibromas are indistinguishable from those of tuberous sclerosis (a TSC1/TSC2–mTOR disorder), so it is the company they keep — the endocrine tumors — that points to MEN1. MEN2A (Sipple syndrome) is the oncogene mirror — a gain-of-function variant in the RET receptor tyrosine kinase causing medullary thyroid carcinoma, pheochromocytoma, and parathyroid disease. Its skin clue is cutaneous lichen amyloidosis: pruritic, rippled, hyperpigmented plaques over the upper back and interscapular region that cluster with RET codon-634 variants — in effect a visible marker of medullary-thyroid-cancer risk.[3] Its more aggressive sibling, MEN2B, follows the RET M918T variant and is recognizable on exam by mucosal neuromas of the lips and tongue and a marfanoid habitus, with early, aggressive medullary thyroid carcinoma and pheochromocytoma but, unlike MEN2A, no hyperparathyroidism.[3]

INHERITED ENDOCRINE SYNDROMES & THE SKIN • CONTINUED

THE CYCLIC-AMP/PKA SYNDROMES

Carney complex is the cleanest illustration of the cAMP theme. Inactivating variants in *PRKARIA* — the $R1\alpha$ regulatory subunit of PKA, found in roughly 70% of cases — release the PKA catalytic subunits from cAMP control, so the kinase fires constitutively. The result is a distinctive skin picture (perioral and eyelid lentiginos, multiple blue nevi, and cutaneous myxomas) paired with endocrine overactivity: ACTH-independent Cushing from primary pigmented nodular adrenocortical disease (PPNAD), acromegaly, and thyroid tumors.[4] The same axis explains two *GNAS* ($Gs\alpha$) disorders at opposite poles: McCune–Albright syndrome (a mosaic activating variant) gives large, jagged “coast-of-Maine” cafe-au-lait macules with fibrous dysplasia and precocious puberty, while pseudohypoparathyroidism (a loss-of-function variant causing PTH resistance) produces Albright hereditary osteodystrophy — round face, short stature, brachydactyly, and subcutaneous ossifications.

RAS AND PTEN: NF1 AND COWDEN

Neurofibromatosis type 1 stems from loss of neurofibromin, a negative regulator of RAS; the cutaneous hallmarks — cafe-au-lait macules, dermal neurofibromas, and axillary or inguinal freckling — carry an endocrine tail, most importantly pheochromocytoma.[1] Cowden syndrome (PTEN loss) is almost universally cutaneous — facial trichilemmomas, oral papillomatosis (cobblestoning), and acral keratoses — and carries a high lifetime risk of thyroid (often follicular) carcinoma and goiter.

AUTOIMMUNE POLYENDOCRINE SYNDROMES

Where the syndromes above are neoplastic, the autoimmune polyendocrine syndromes are failures of self-tolerance. APS-1 (APECED), caused by recessive *AIRE* variants, presents early with chronic mucocutaneous candidiasis, hypoparathyroidism, and primary adrenal insufficiency, alongside vitiligo, alopecia areata, and nail or enamel dystrophy. APS-2 and APS-3 are polygenic and HLA-linked, pairing autoimmune thyroid disease and type 1 diabetes with the same dermatologic markers — vitiligo and alopecia areata.[5]

INHERITED ENDOCRINE SYNDROMES & THE SKIN • CONTINUED

INHERITED ADRENAL AND METABOLIC DISEASE

The adrenal failure of APS is autoimmune, but primary adrenal insufficiency is just as often inherited — congenital adrenal hyperplasia (21-hydroxylase deficiency, CYP21A2), X-linked adrenoleukodystrophy (ABCD1), and the ACTH-resistance syndromes (familial glucocorticoid deficiency, Allgrove). Whatever the cause, the skin sign is the same and is mechanistically elegant: loss of cortisol feedback drives POMC-derived ACTH and α -MSH, so melanocortin signaling rises and the patient develops hyperpigmentation — darkest in palmar creases, scars, and mucosa — with congenital adrenal hyperplasia adding androgen-driven hirsutism and acne.[1] A second metabolic-skin pairing is familial partial lipodystrophy (LMNA, PPARG): selective loss of limb and gluteal subcutaneous fat gives a deceptively muscular build, while the redirected metabolic load produces severe insulin resistance, early diabetes, and florid acanthosis nigricans — the same cutaneous readout of hyperinsulinemia seen in the receptor defects below.[1]

CUTANEOUS CLUES AT A GLANCE

SYNDROME • GENE • PATHWAY • KEY SKIN SIGNS • ENDOCRINE FEATURE

MEN1 (Wermer)	MEN1 • menin (suppressor) • Facial angiofibromas, collagenomas, lipomas • Parathyroid, pituitary, pancreatic NETs
MEN2A (Sipple)	RET • RTK (oncogene) • Cutaneous lichen amyloidosis (interscapular) • Medullary thyroid CA, pheo, parathyroid
MEN2B (Wagenmann–Froboese)	RET (M918T) • oncogene • Mucosal neuromas (lips, tongue), marfanoid habitus • Aggressive medullary thyroid CA, pheo (no PHPT)
Carney complex	PRKARIA • cAMP/PKA • Lentigines, blue nevi, myxomas • PPNAD/Cushing, acromegaly, thyroid
McCune–Albright	GNAS (activating) • cAMP • “Coast-of-Maine” cafe-au-lait • Precocious puberty, fibrous dysplasia, acromegaly
Pseudohypoparathyroidism	GNAS (loss) • cAMP • Albright osteodystrophy; round face, brachydactyly, subcutaneous ossification • PTH resistance, hypocalcemia
Neurofibromatosis 1	NF1 • neurofibromin / RAS • Cafe-au-lait, neurofibromas, axillary freckling • Pheochromocytoma
Cowden	PTEN • PI3K/AKT • Trichilemmomas, oral papillomatosis, acral keratoses • Thyroid carcinoma, goiter

INHERITED ENDOCRINE SYNDROMES & THE SKIN · CONTINUED

APS-1 (APECED)	AIRE · Mucocutaneous candidiasis, vitiligo, alopecia · Hypoparathyroidism, primary adrenal insufficiency
APS-2 / APS-3	HLA / polygenic · Vitiligo, alopecia areata · Primary adrenal insufficiency, autoimmune thyroid, type 1 diabetes
Inherited primary adrenal insufficiency	CYP21A2, ABCD1, MC2R · POMC-ACTH · Hyperpigmentation; CAH adds hirsutism, acne · Cortisol deficiency (± androgen excess)
Familial insulin resistance (type A)	INSR · Severe acanthosis nigricans, skin tags · Hyperinsulinemia, hyperandrogenism
Familial partial lipodystrophy	LMNA, PPARG · Acanthosis nigricans; lipotrophic limbs · Insulin-resistant diabetes, hypertriglyceridemia

CLINICAL DATA

MEN1 skin	facial angiofibromas in ~88% and collagenomas in ~72% — cutaneous clues that often precede the endocrine diagnosis.[2]
Carney	PRKAR1A inactivation (~70% of cases) releases PKA from cAMP control; lentigines and myxomas flag PPNAD-driven Cushing and acromegaly.[4]
MEN2A	cutaneous lichen amyloidosis clusters with RET codon-634 variants — a skin marker of medullary-thyroid-cancer risk.[3]

THREE THINGS TO REMEMBER

1. Read the skin first: angiofibromas and collagenomas (MEN1), lichen amyloidosis (MEN2A), and lentigines with myxomas (Carney) can all precede the endocrine diagnosis.
2. Two mechanistic buckets explain most of it: tumor-suppressor and oncogene pathways (menin, RET, NF1-RAS, PTEN) and the cAMP-PKA axis (PRKAR1A, GNAS) — the wiring behind Cushing, acromegaly, and PTH resistance.
3. A pathognomonic skin sign is a referral, not a cosmetic problem: pair the finding with targeted endocrine and genetic work-up.

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A NOTE TO THE READER

This guide reflects the state of the evidence as the author reads it at the time of writing. Aesthetic and regenerative endocrinology moves quickly; readers are encouraged to verify citations, approvals, and guidelines against primary sources, and to treat the investigational interventions described here as exactly that.

About this guide

Aesthetic & Regenerative Endocrinology — A Pocket Guide is an educational reference prepared for healthcare providers. It distills a fast-moving field into a compact, evidence-rated form: the physiology of endocrine aging, the organs of aging, the interventions that optimize them, and the regenerative frontier of endocrine tissue replacement — with a framework for responsible practice.

Every chapter has been written against primary literature, landmark trials, and authoritative reviews, with citations provided per chapter so that readers can go to the source. Strength-of-evidence ratings reflect the state of published evidence and the distinction between established clinical use, useful adjuncts, and investigational claims. Regulatory status is noted where it bears on practice.

A LIVING REFERENCE

This field evolves quickly — trials report, approvals change, and compounding rules are revised. Treat this edition as a snapshot. Verify testing recommendations, drug approvals, compounding status, and every citation against current primary sources before acting.

SCOPE & DISCLAIMER

This guide is educational and is not medical advice for any individual patient. It does not recommend unapproved peptides, bioregulators, or unregulated cell therapies for routine clinical use, and it does not replace specialist input, formal guidelines, or individualized clinical judgment.

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